Airfoil: first six generalized Ritz modes

Four methods | three parameter frames | velocity + streamlines and vorticity

Each following page compares one mode index across all four methods using the stored mass-orthogonal Ritz basis.

Upper frame | beta=0.50, alpha=+10 deg

Method	Epoch	lambda 1	lambda 2	lambda 3	lambda 4	lambda 5	lambda 6	M-orth defect
FP16 V100	1000	0.80260	1.13368	1.57917	2.03229	2.39923	2.58762	2.68e-14
FP16 V50/W50	650	0.55931	0.94240	1.32807	1.59642	2.09740	2.25614	4.70e-13
FP64 V100	5000	0.61319	0.95881	1.39081	1.65832	2.15912	2.37648	9.18e-13
FP16 W100	5000	0.57949	0.92927	1.38196	1.61600	2.08951	2.29383	1.17e-12

Middle frame | beta=0.85, alpha=0 deg

Method	Epoch	lambda 1	lambda 2	lambda 3	lambda 4	lambda 5	lambda 6	M-orth defect
FP16 V100	1000	0.79602	1.11231	1.53754	1.88026	2.34874	2.52172	1.79e-14
FP16 V50/W50	650	0.56033	0.94345	1.33370	1.59834	2.17510	2.28169	4.49e-13
FP64 V100	5000	0.60581	0.94975	1.35920	1.62521	2.19582	2.32233	2.40e-13
FP16 W100	5000	0.58818	0.93354	1.40940	1.65444	2.17101	2.30261	4.39e-12

Lower frame | beta=0.50, alpha=-10 deg

Method	Epoch	lambda 1	lambda 2	lambda 3	lambda 4	lambda 5	lambda 6	M-orth defect
FP16 V100	1000	0.84762	1.12627	1.57887	1.95559	2.32736	2.59535	7.55e-14
FP16 V50/W50	650	0.56047	0.93972	1.33139	1.58939	2.09645	2.29104	5.51e-13
FP64 V100	5000	0.64131	0.96597	1.38753	1.63111	2.18016	2.36166	6.71e-13
FP16 W100	5000	0.58002	0.92982	1.38974	1.66097	2.11859	2.28296	4.27e-13

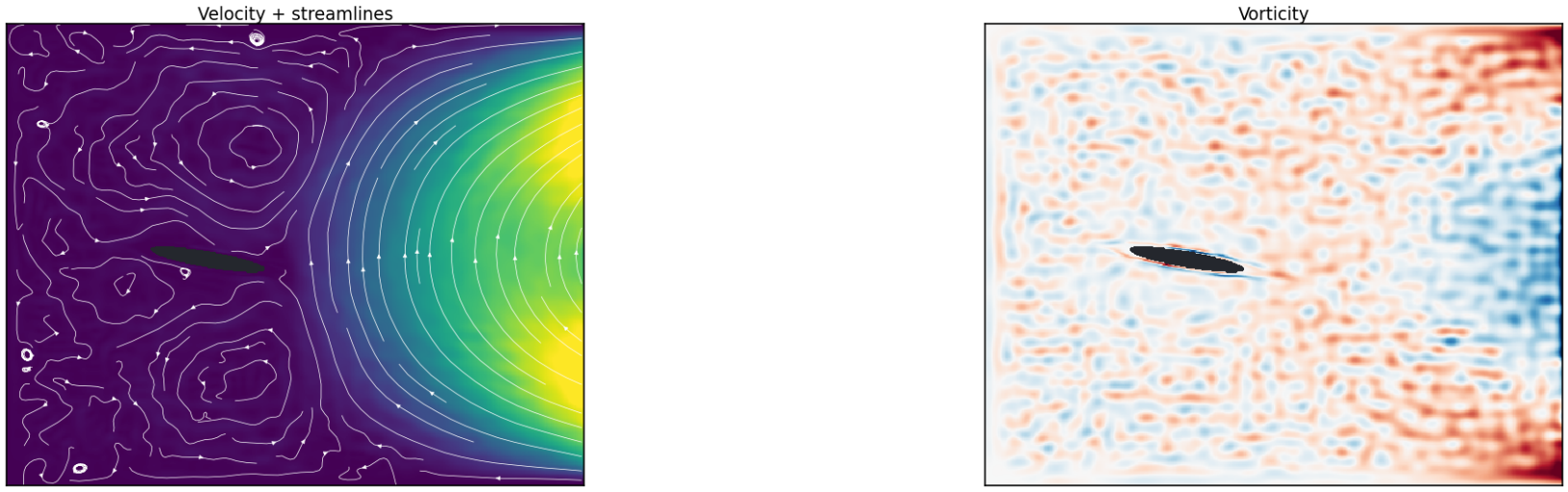
All values and images are read directly from the completed three-frame evaluation artifacts.

Upper frame | generalized Ritz mode 1 of 6

beta=0.50, alpha=+10 deg | left: velocity + streamlines | right: vorticity

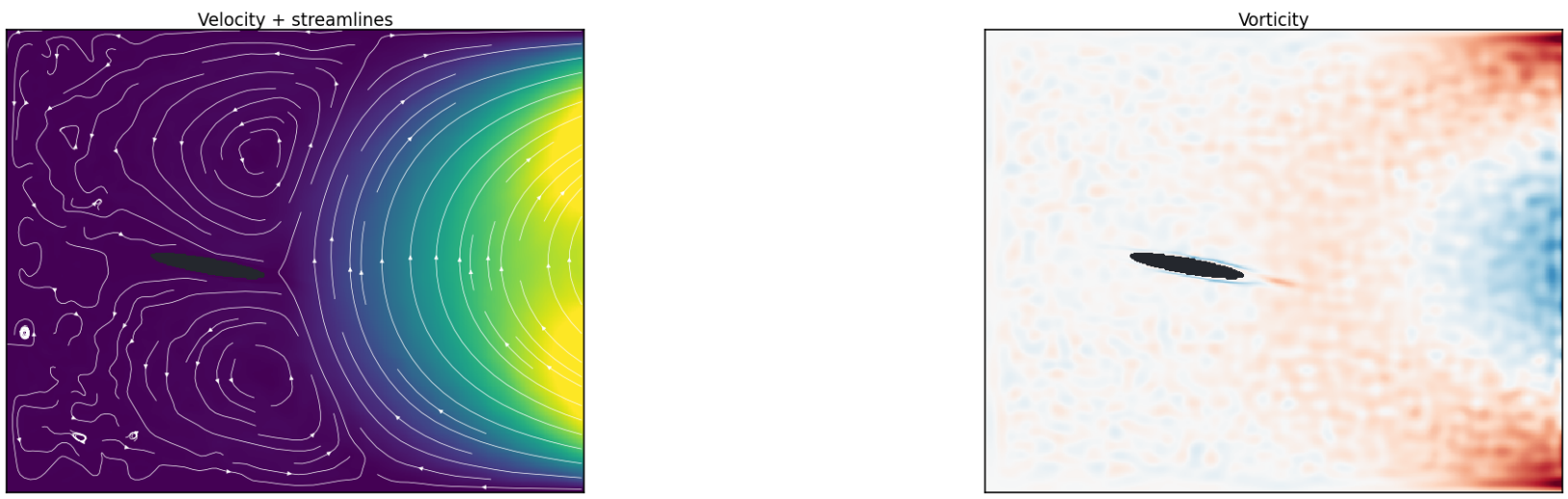
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 1, $\lambda = 8.02605e-01$



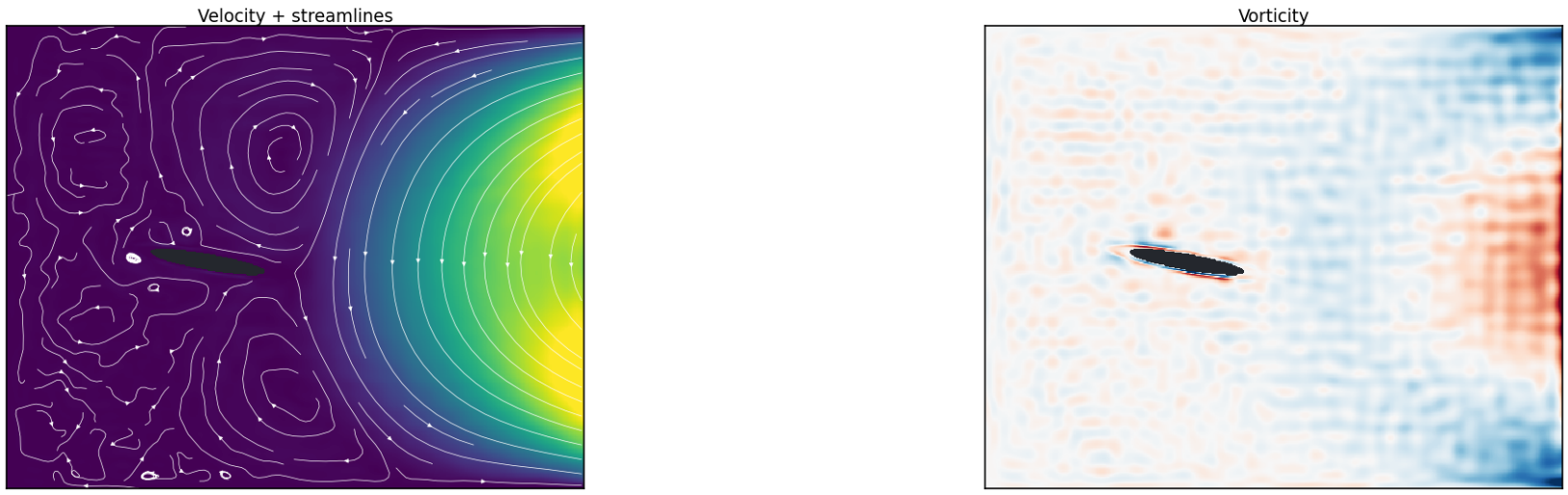
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 1, $\lambda = 5.59313e-01$



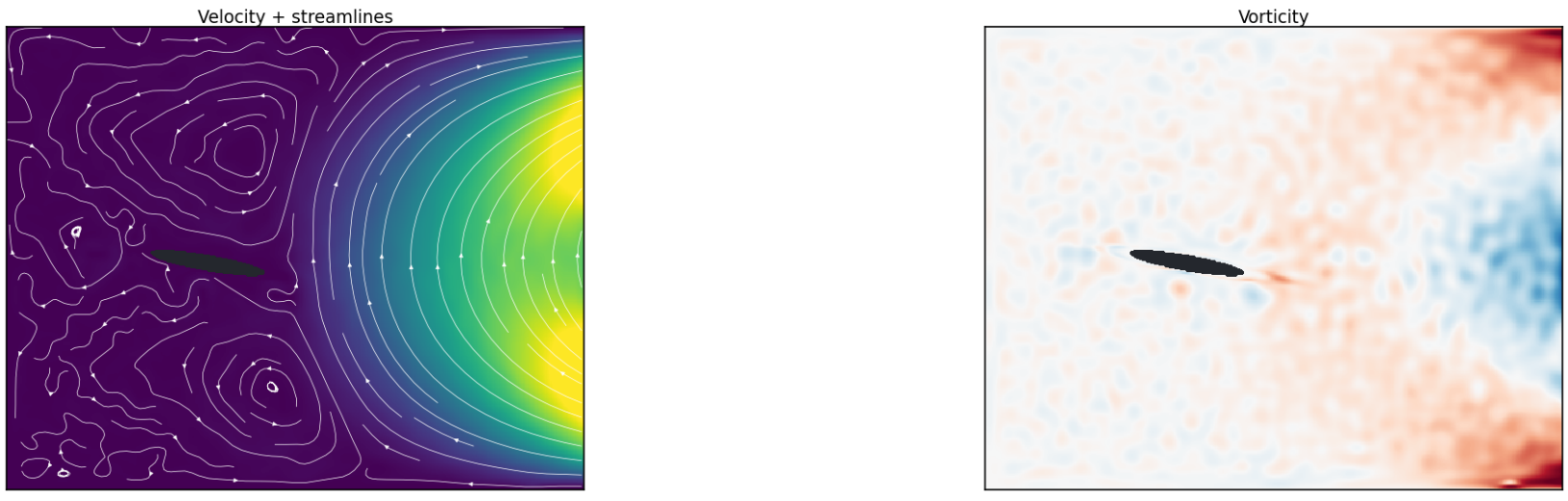
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 1, $\lambda = 6.13195e-01$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 1, $\lambda = 5.79485e-01$

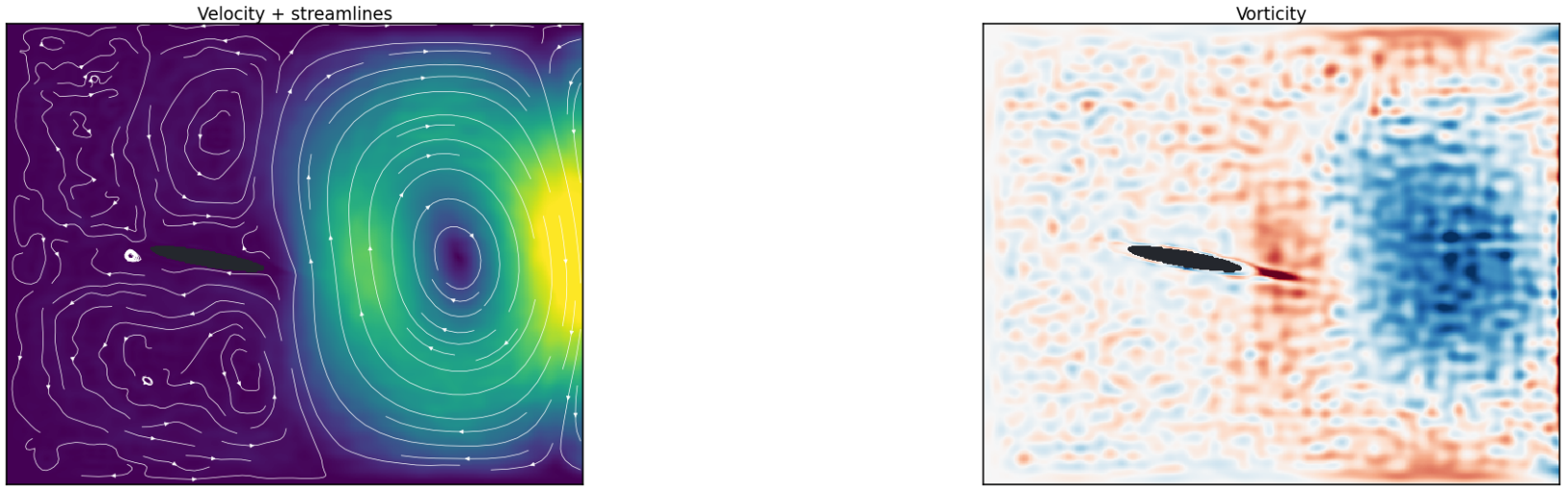


Upper frame | generalized Ritz mode 2 of 6

beta=0.50, alpha=+10 deg | left: velocity + streamlines | right: vorticity

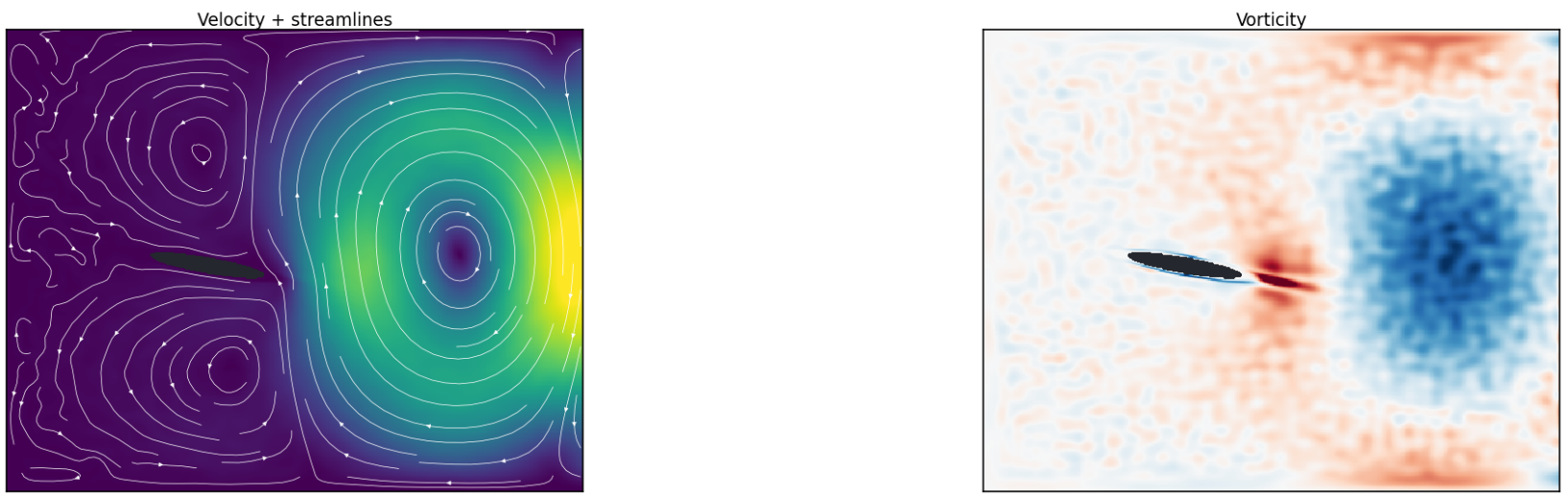
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 2, $\lambda = 1.13368e+00$



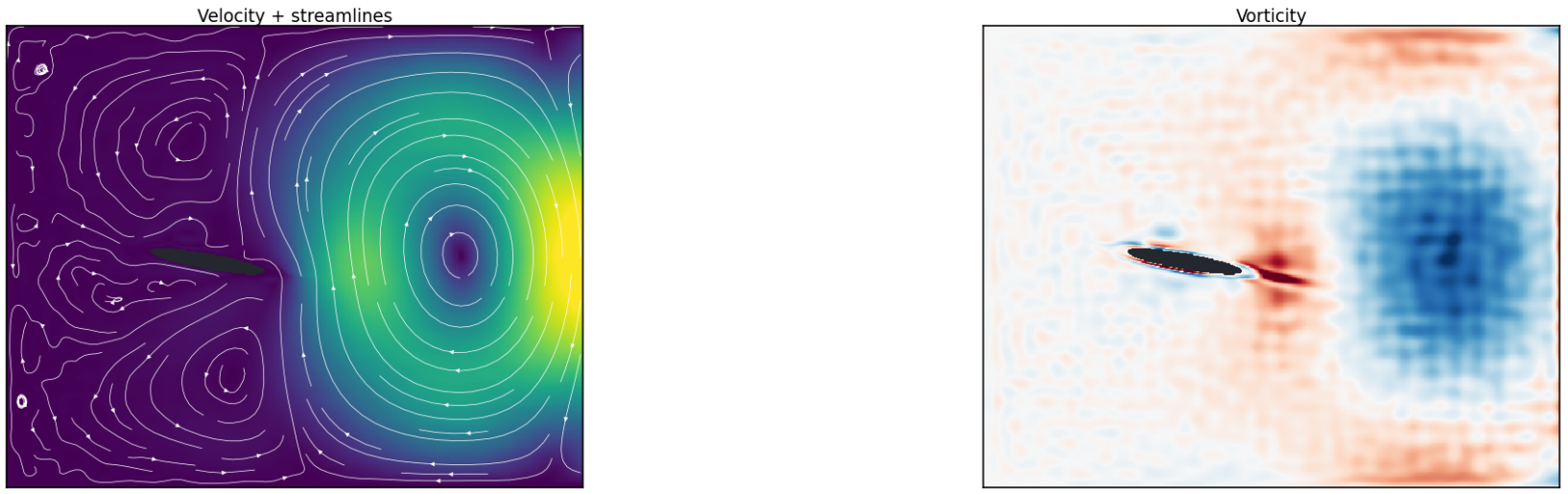
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 2, $\lambda = 9.42404e-01$



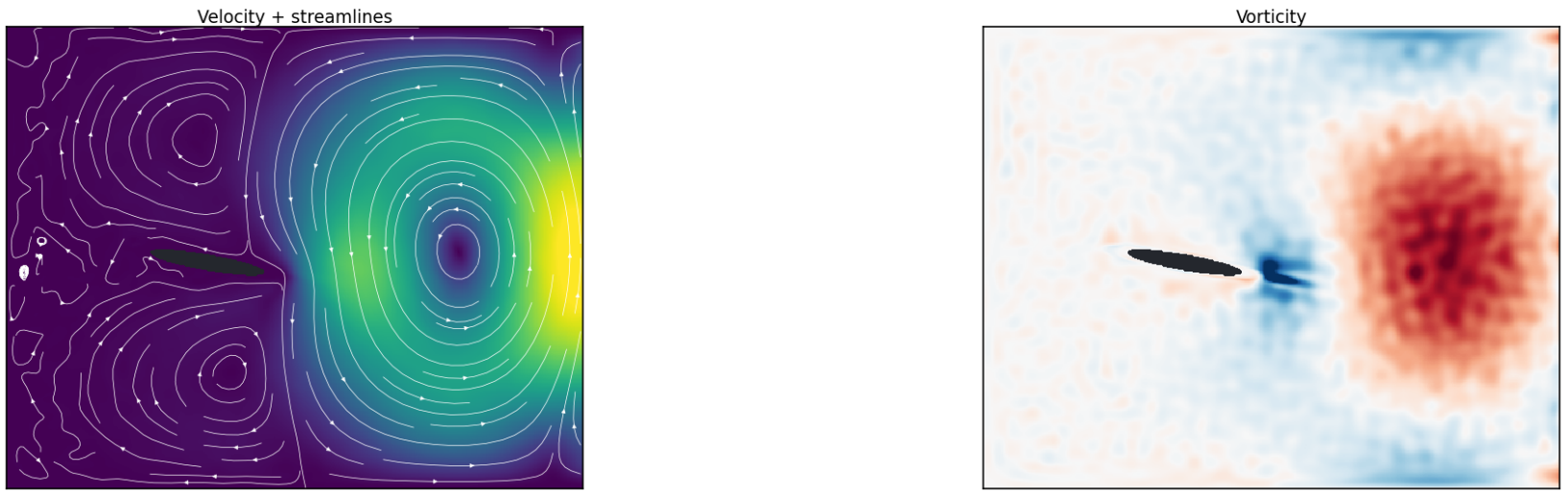
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 2, $\lambda = 9.58814e-01$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 2, $\lambda = 9.29274e-01$

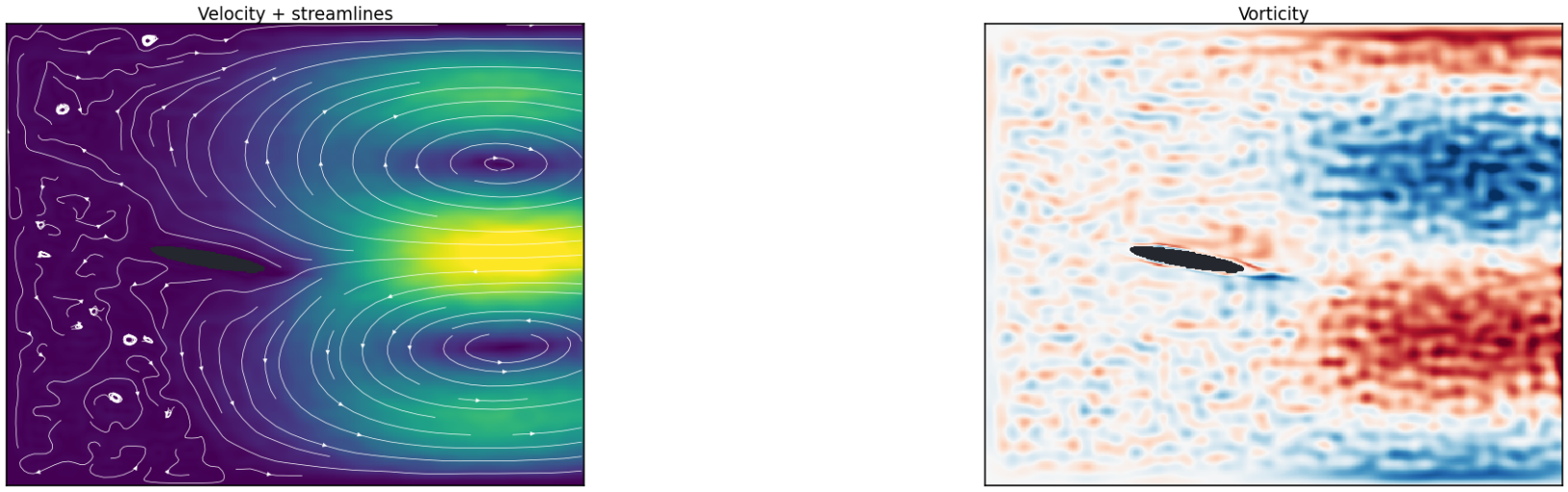


Upper frame | generalized Ritz mode 3 of 6

beta=0.50, alpha=+10 deg | left: velocity + streamlines | right: vorticity

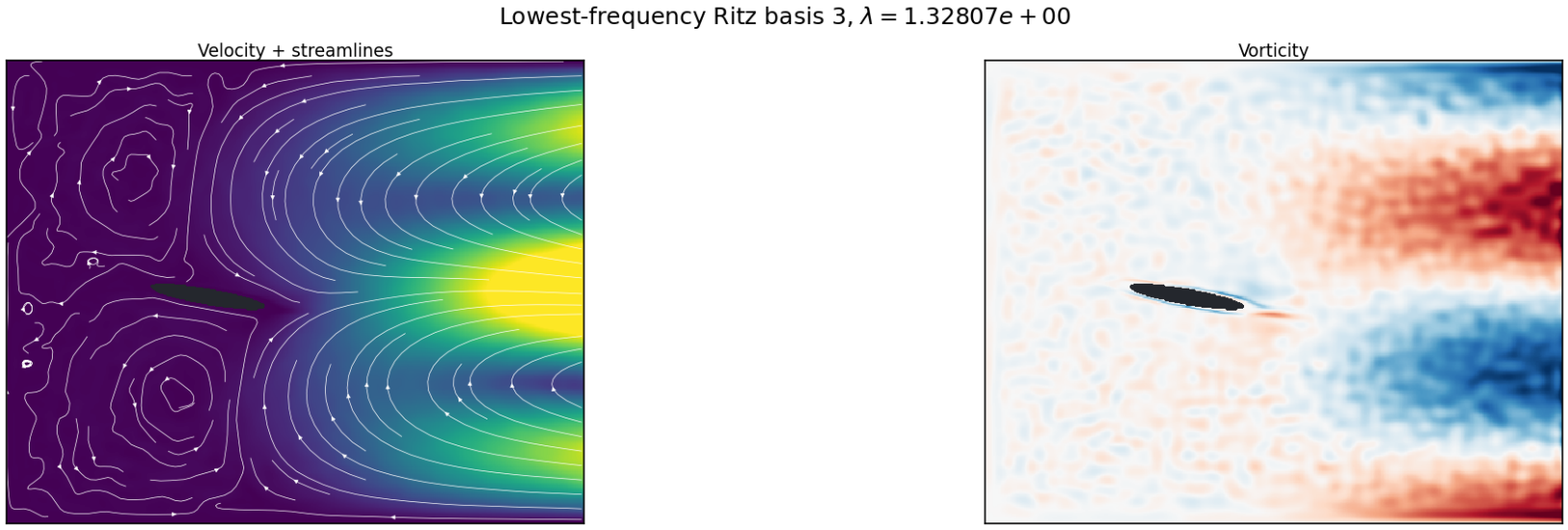
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 3, $\lambda = 1.57917e+00$



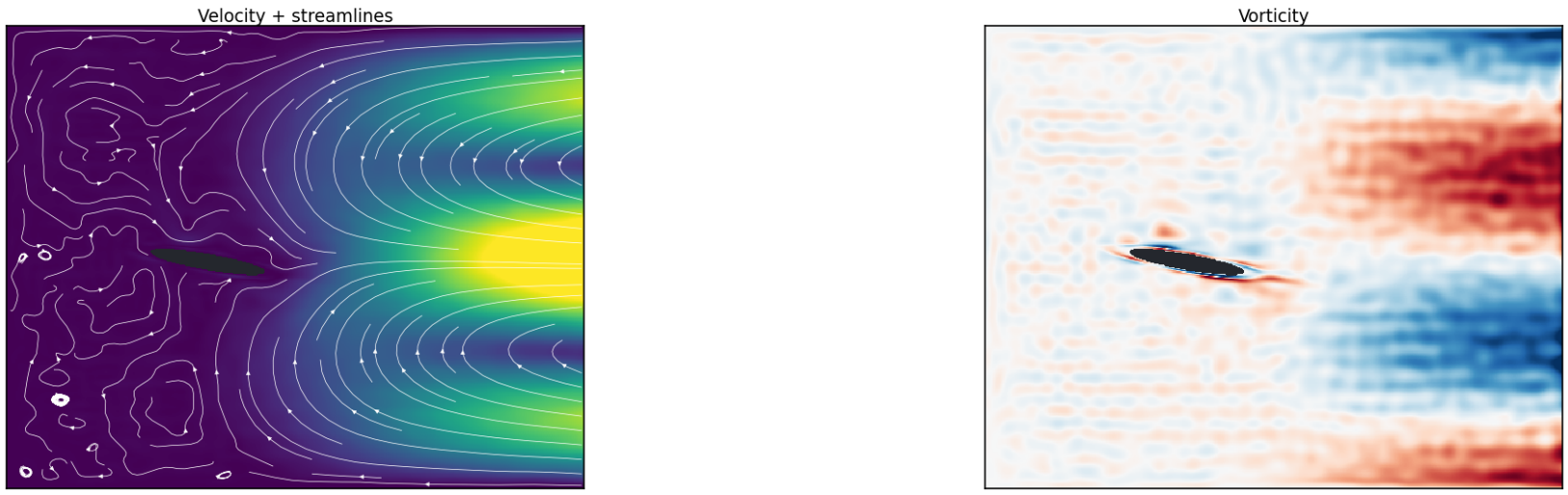
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 3, $\lambda = 1.32807e+00$



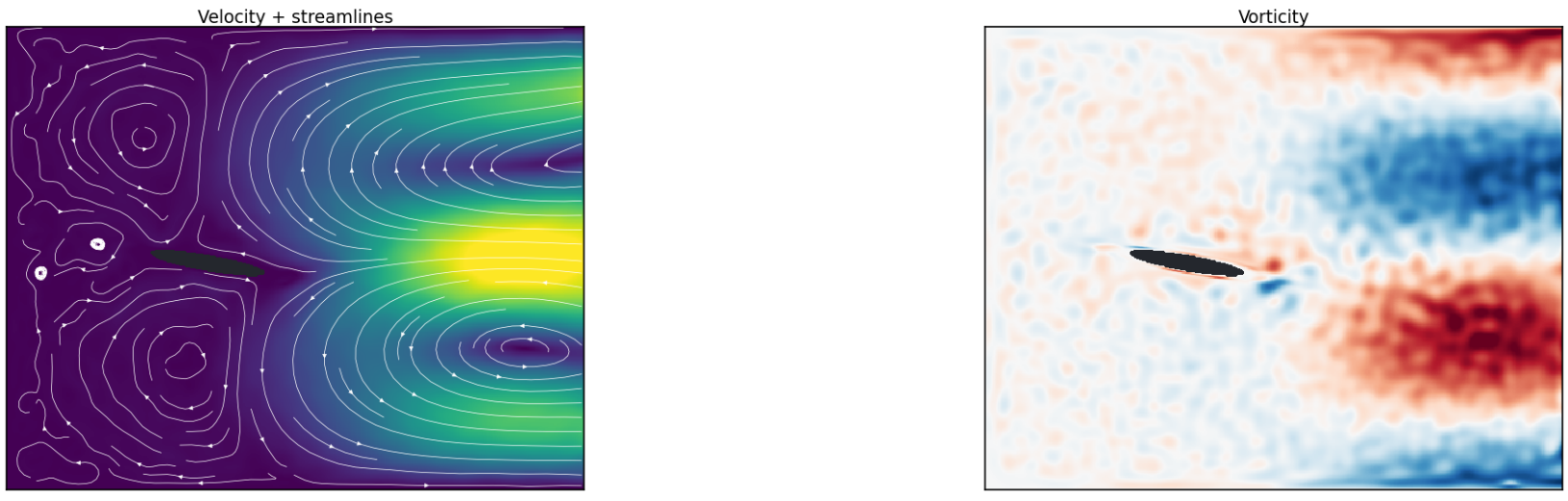
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 3, $\lambda = 1.39081e+00$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

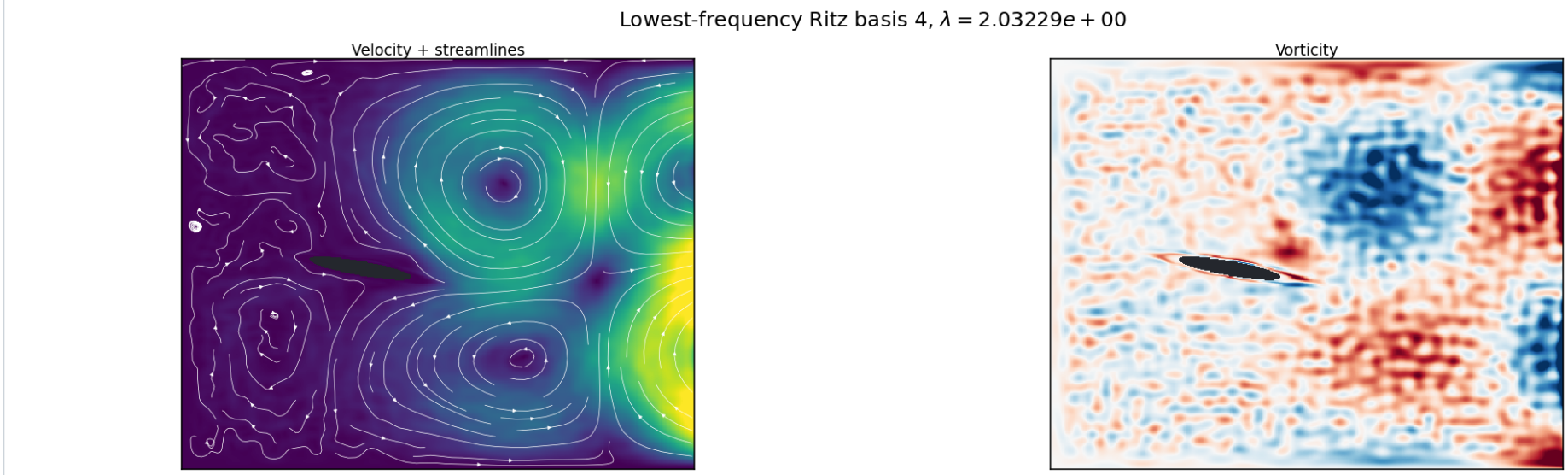
Lowest-frequency Ritz basis 3, $\lambda = 1.38196e+00$



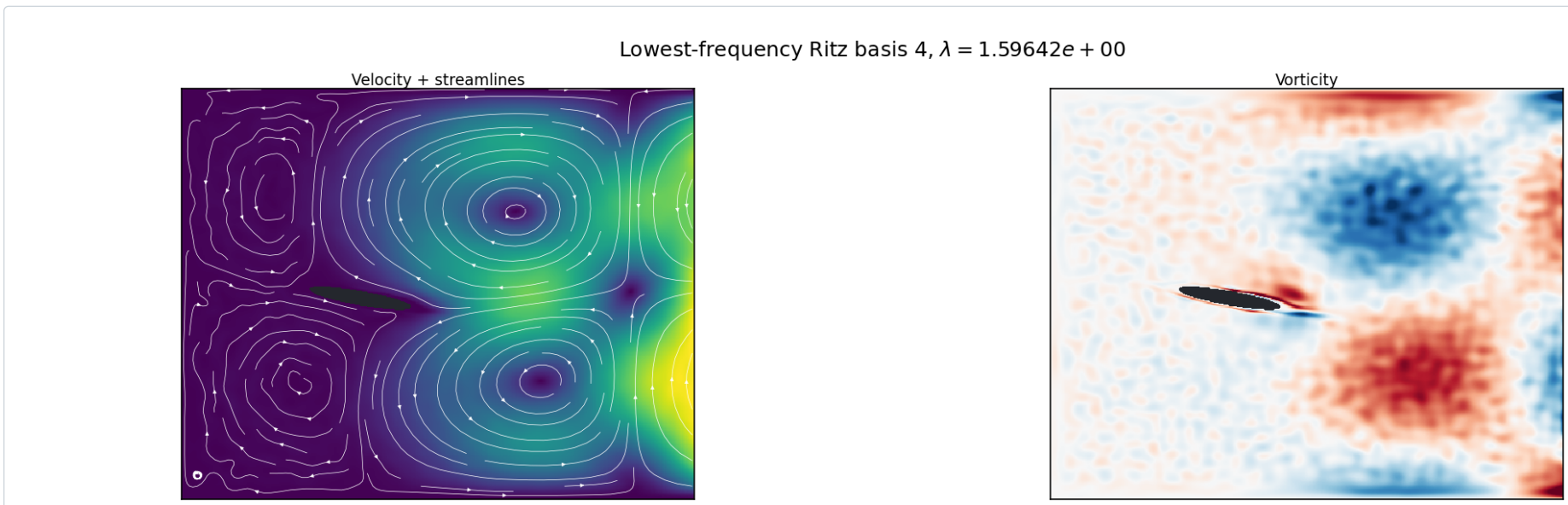
Upper frame | generalized Ritz mode 4 of 6

beta=0.50, alpha=+10 deg | left: velocity + streamlines | right: vorticity

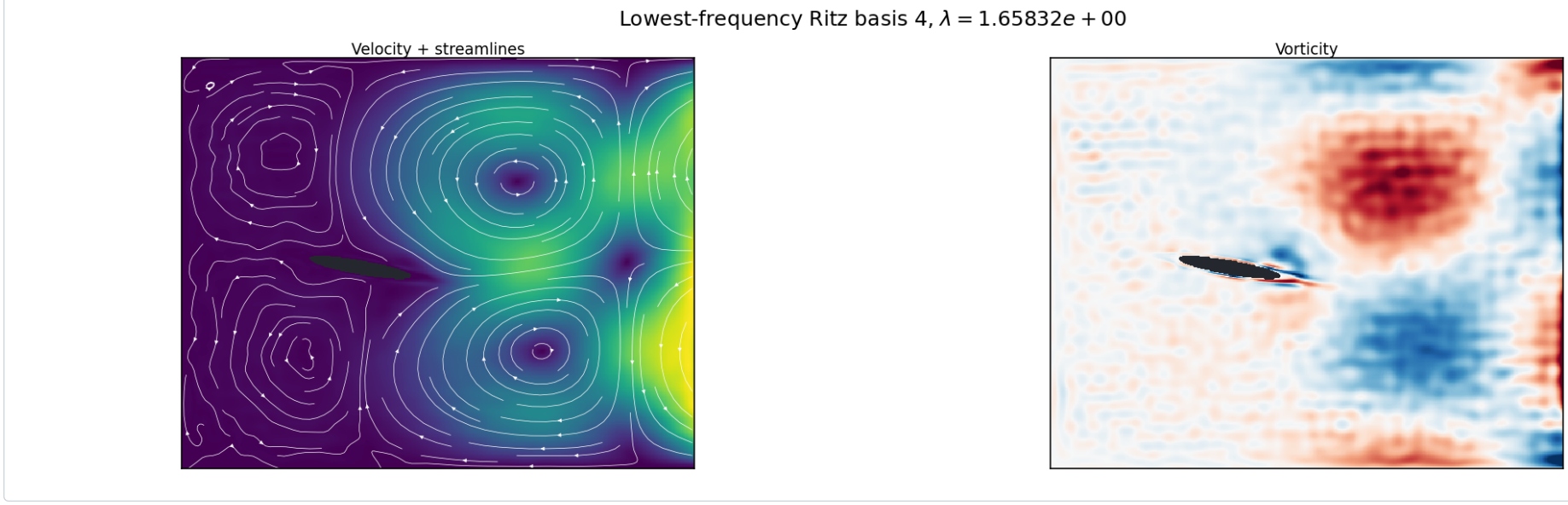
FP16 V100 | epoch 1000 | FP16 velocity 100%



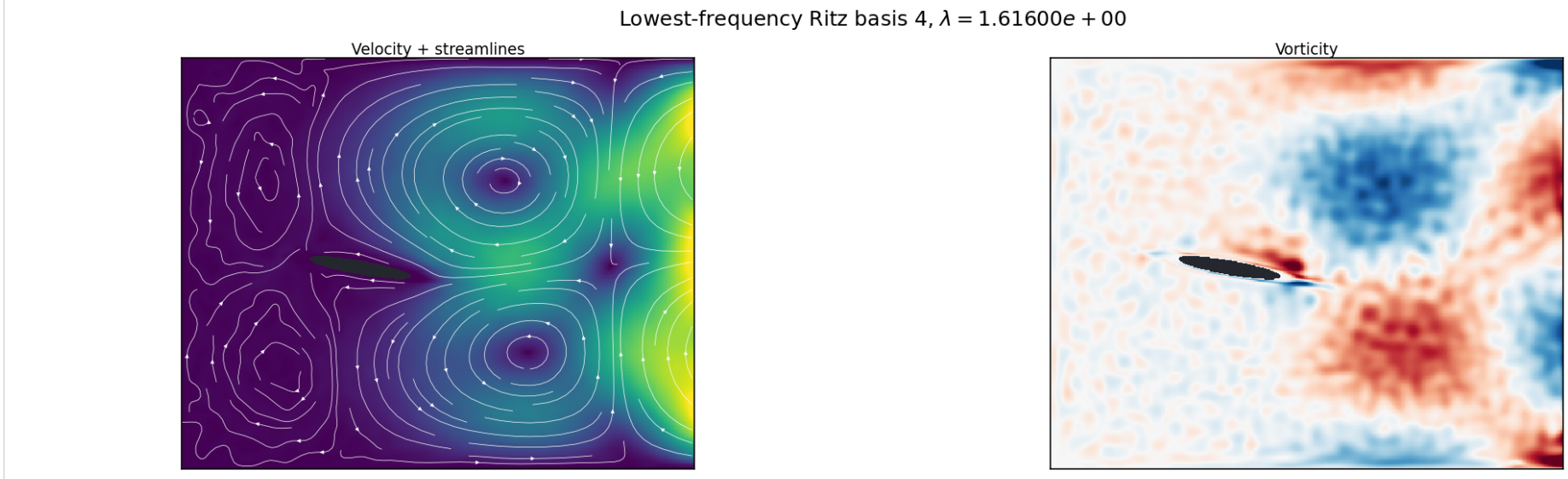
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%



FP64 V100 | epoch 5000 | FP64 velocity 100%



FP16 W100 | epoch 5000 | FP16 vorticity 100%

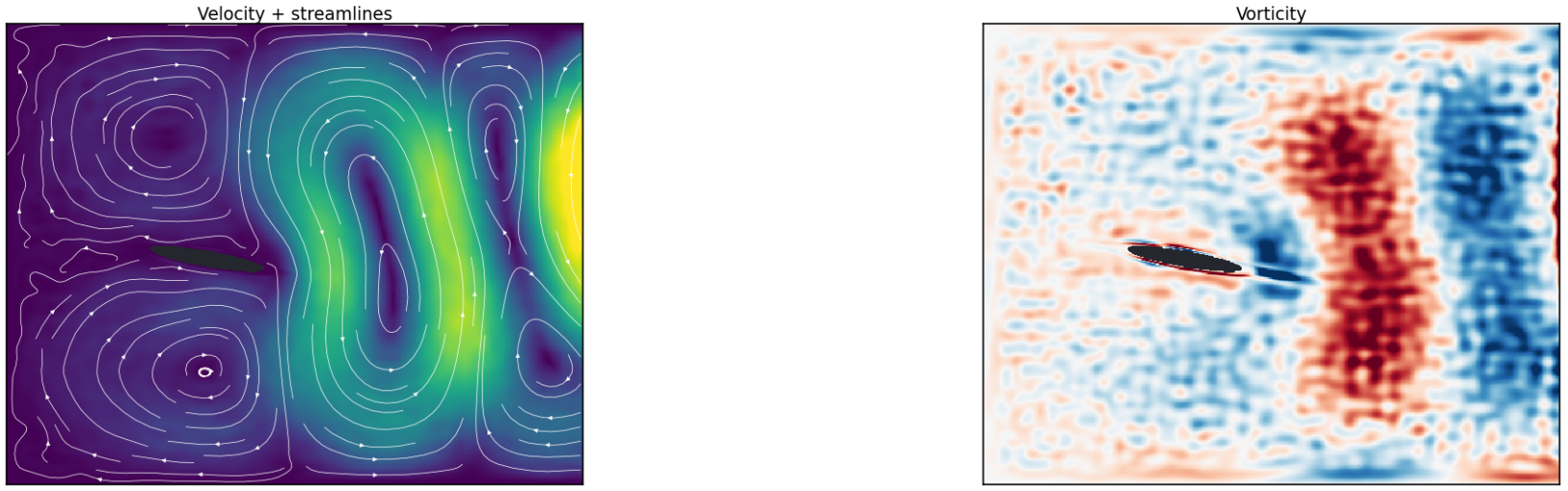


Upper frame | generalized Ritz mode 5 of 6

beta=0.50, alpha=+10 deg | left: velocity + streamlines | right: vorticity

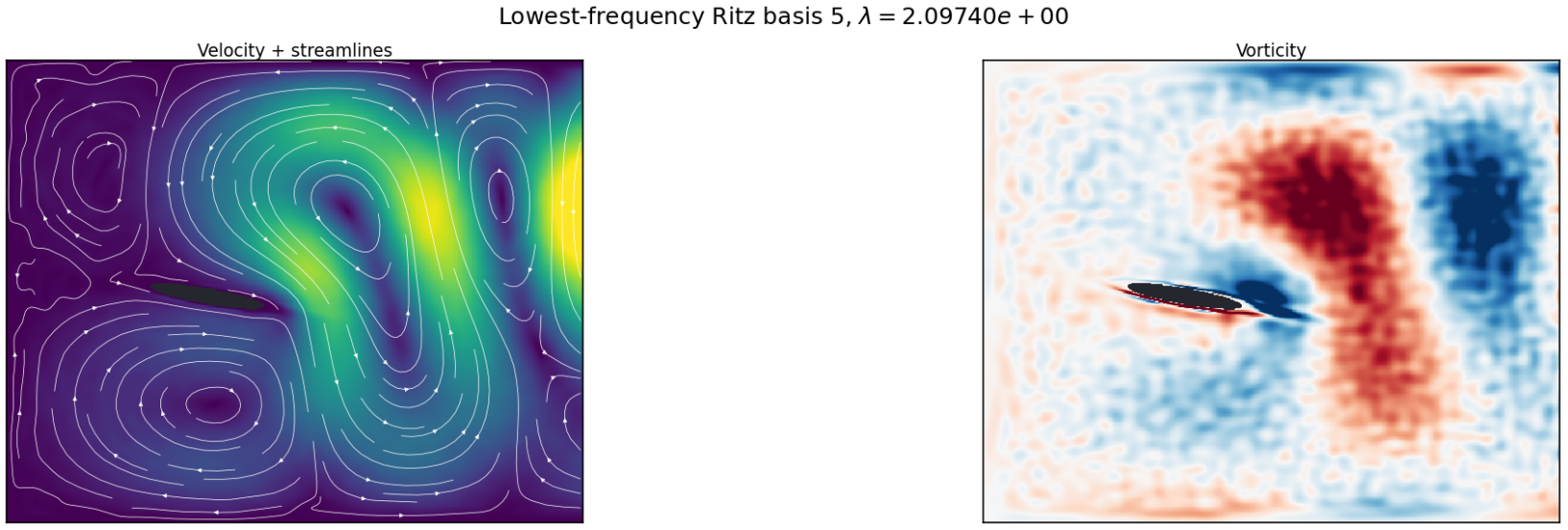
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 5, $\lambda = 2.39923e+00$



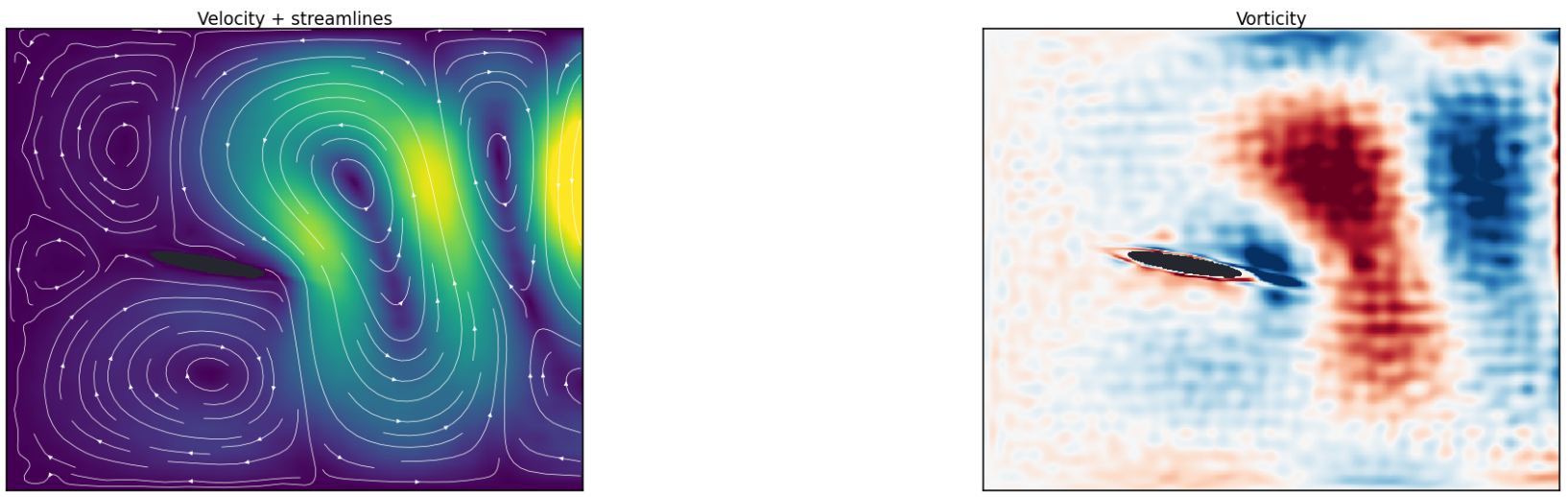
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 5, $\lambda = 2.09740e+00$



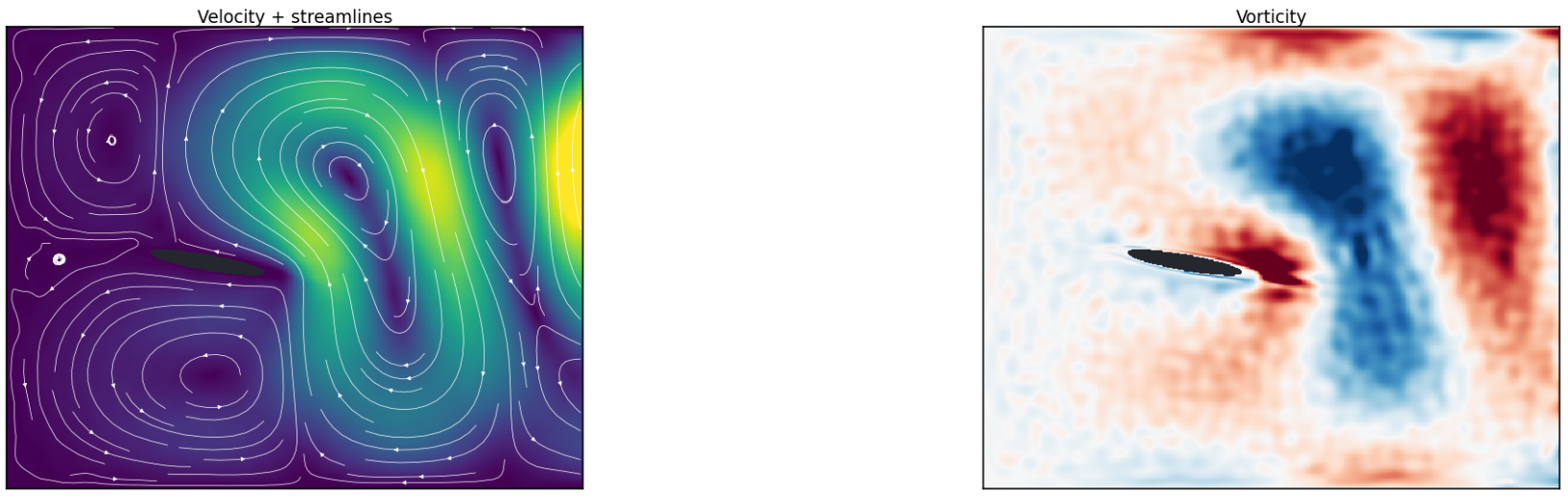
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 5, $\lambda = 2.15912e+00$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 5, $\lambda = 2.08951e+00$

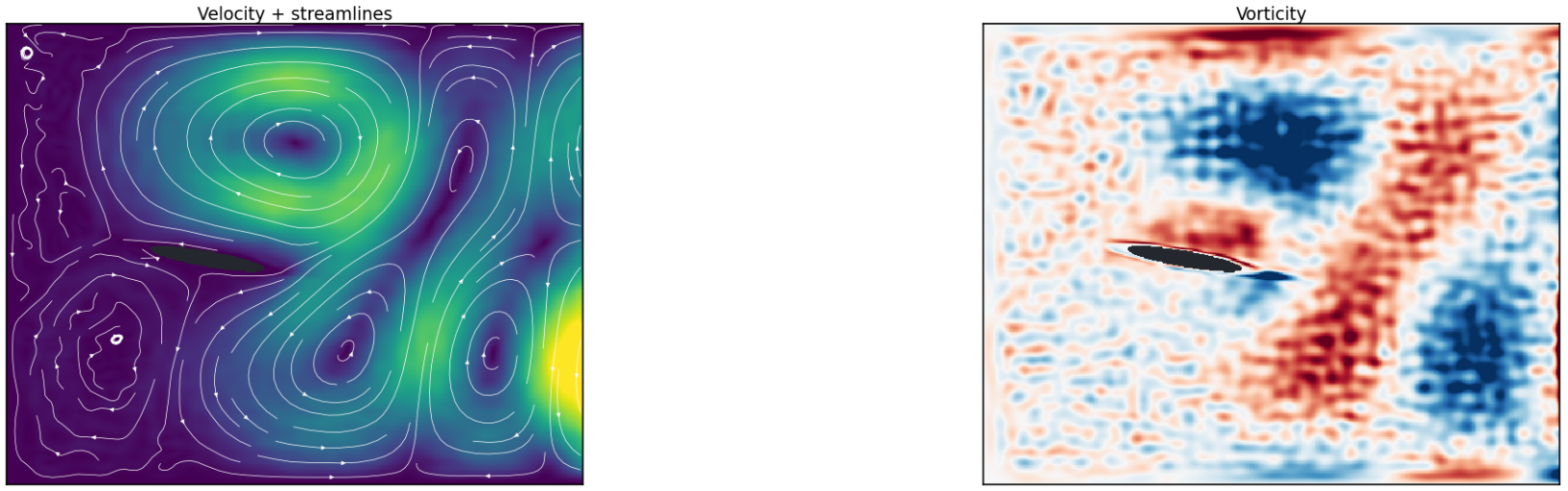


Upper frame | generalized Ritz mode 6 of 6

beta=0.50, alpha=+10 deg | left: velocity + streamlines | right: vorticity

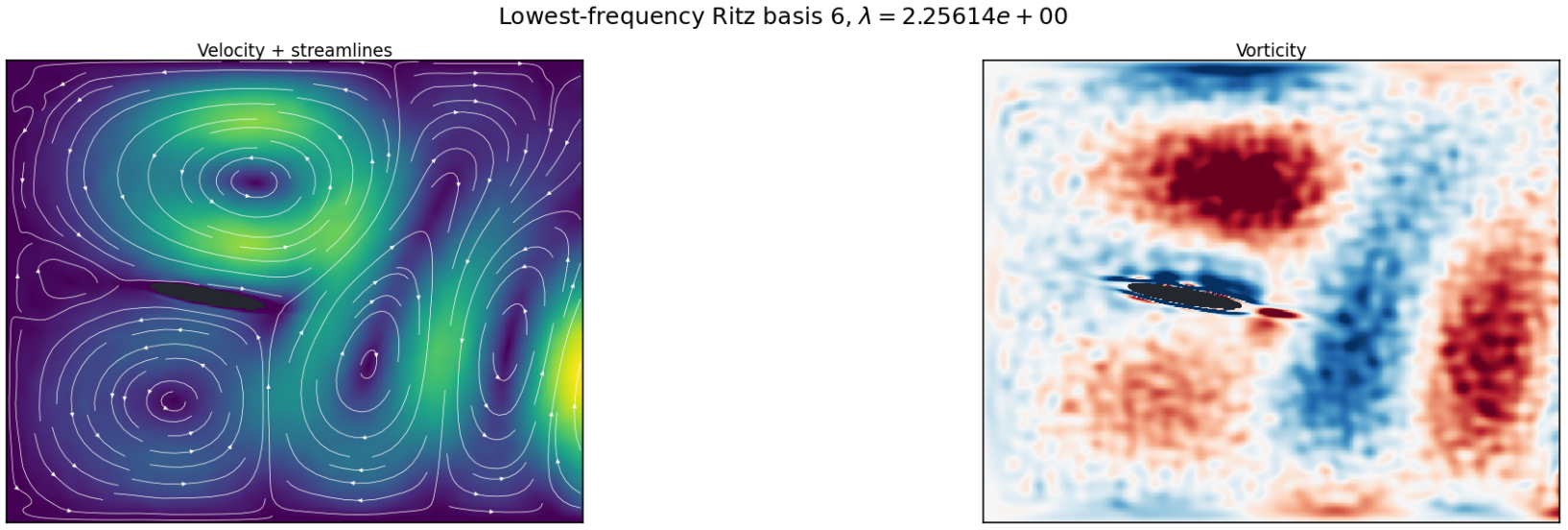
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 6, $\lambda = 2.58762e+00$



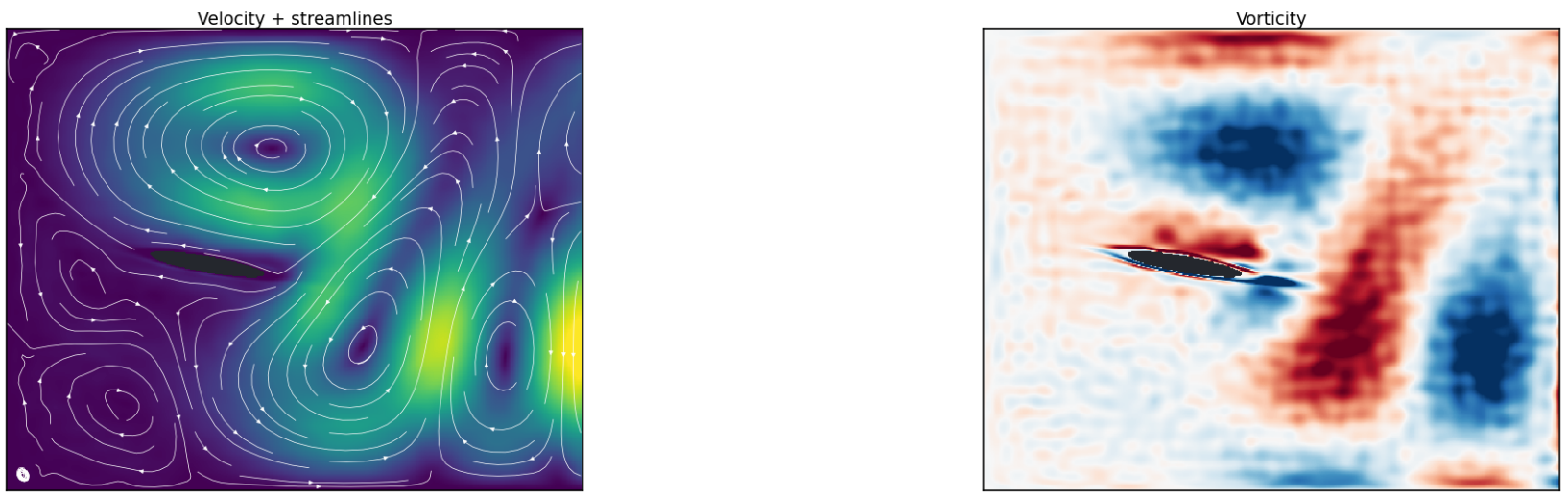
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 6, $\lambda = 2.25614e+00$



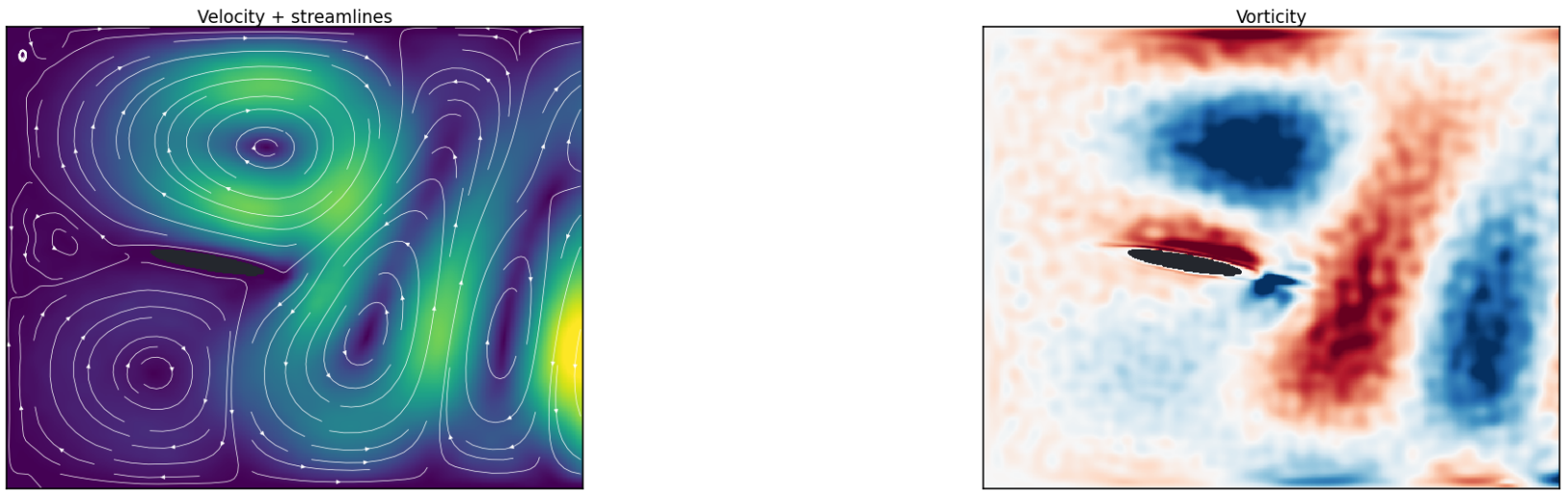
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 6, $\lambda = 2.37648e+00$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 6, $\lambda = 2.29383e+00$

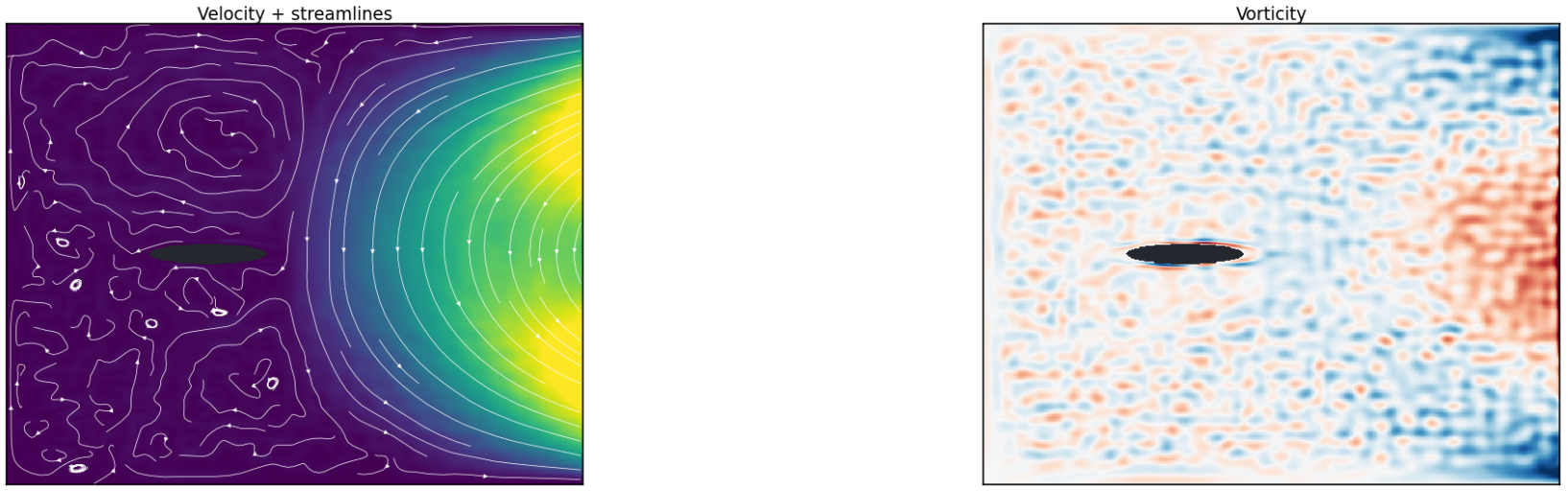


Middle frame | generalized Ritz mode 1 of 6

beta=0.85, alpha=0 deg | left: velocity + streamlines | right: vorticity

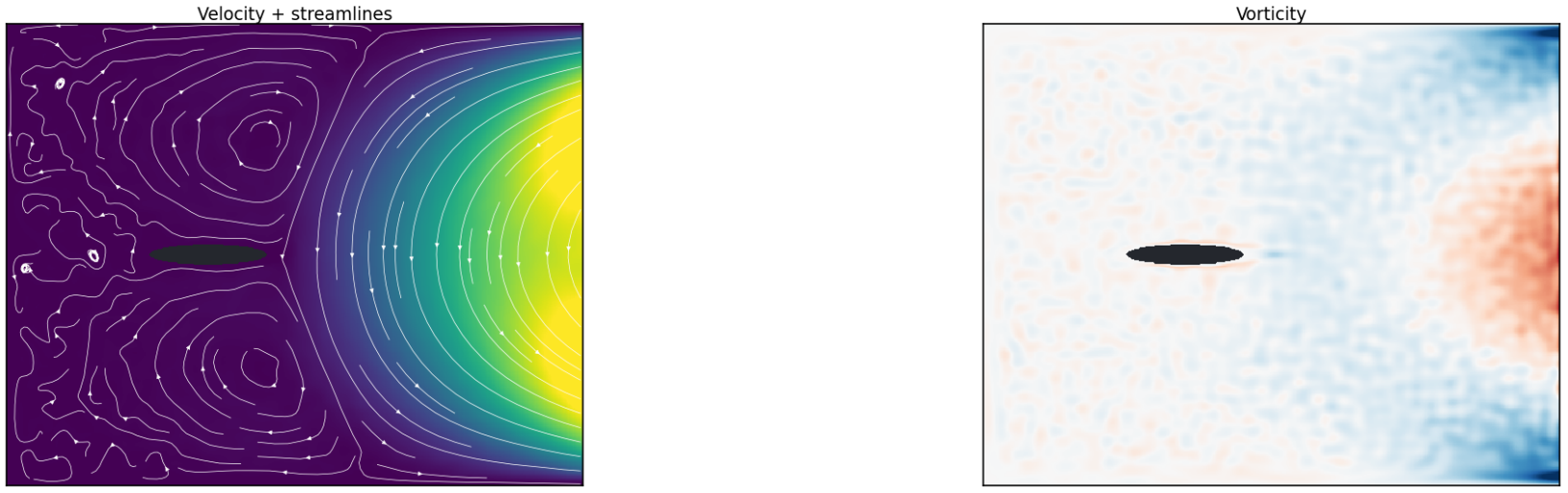
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 1, $\lambda = 7.96017e-01$



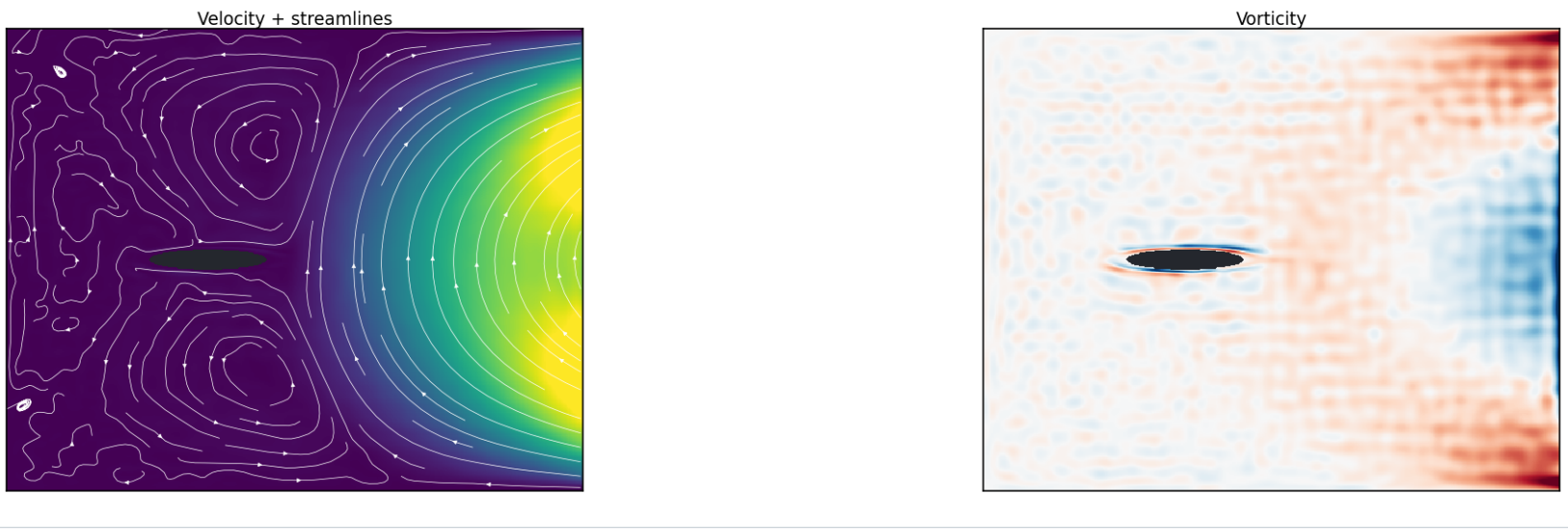
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 1, $\lambda = 5.60325e-01$



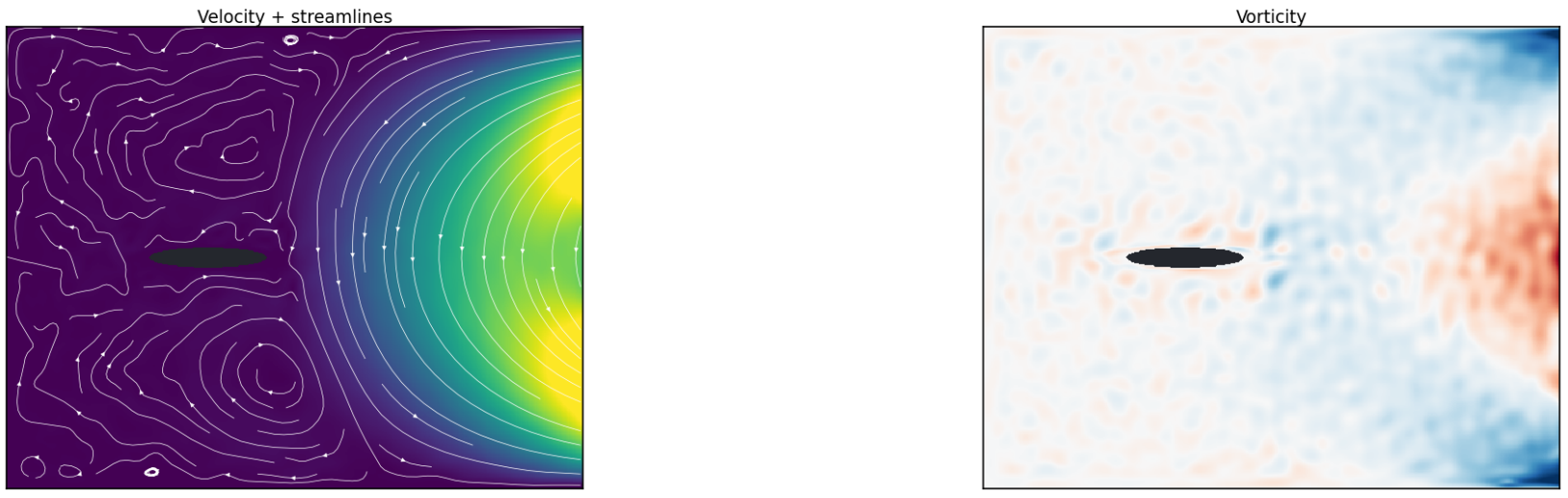
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 1, $\lambda = 6.05814e-01$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 1, $\lambda = 5.88184e-01$

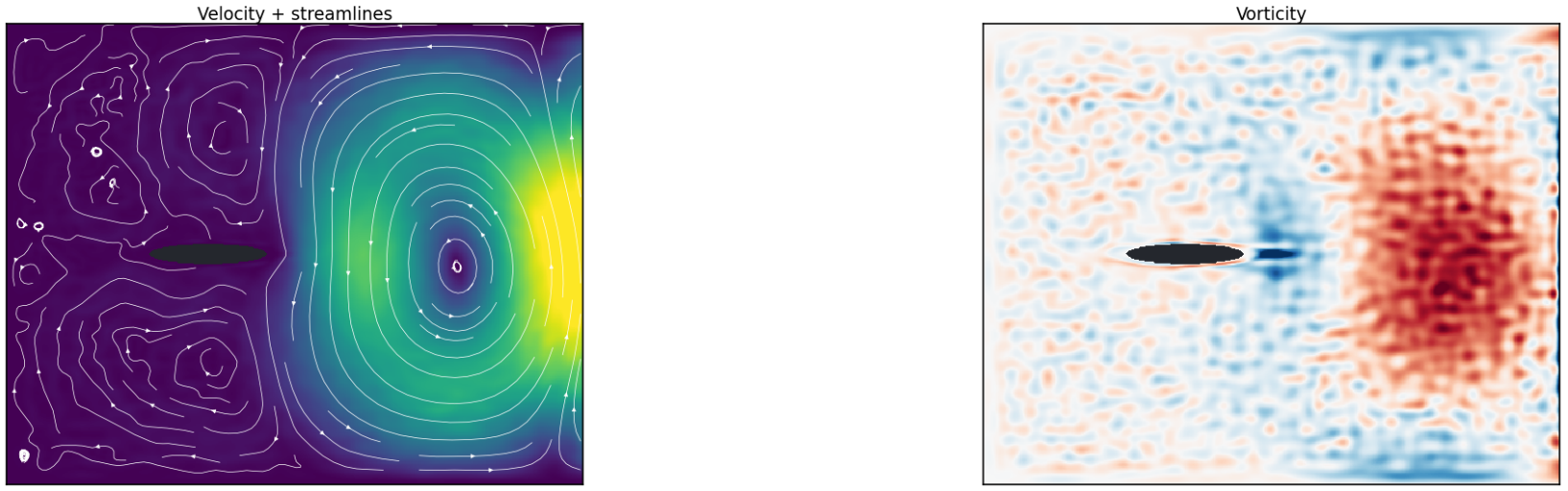


Middle frame | generalized Ritz mode 2 of 6

beta=0.85, alpha=0 deg | left: velocity + streamlines | right: vorticity

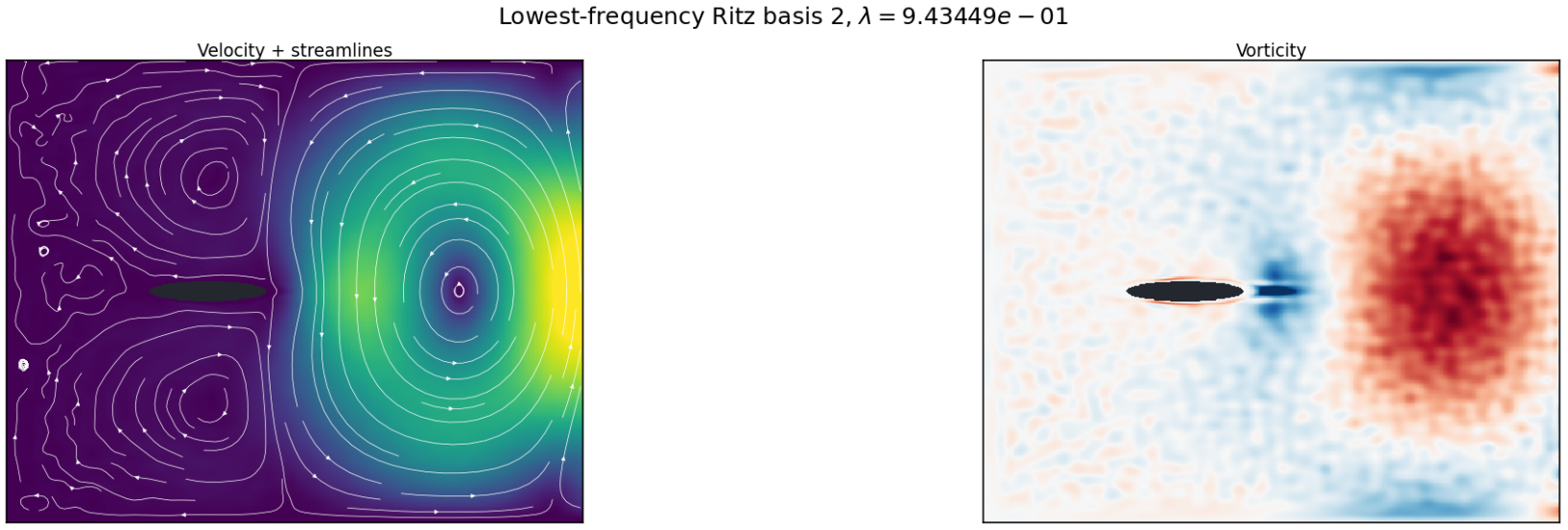
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 2, $\lambda = 1.11231e+00$



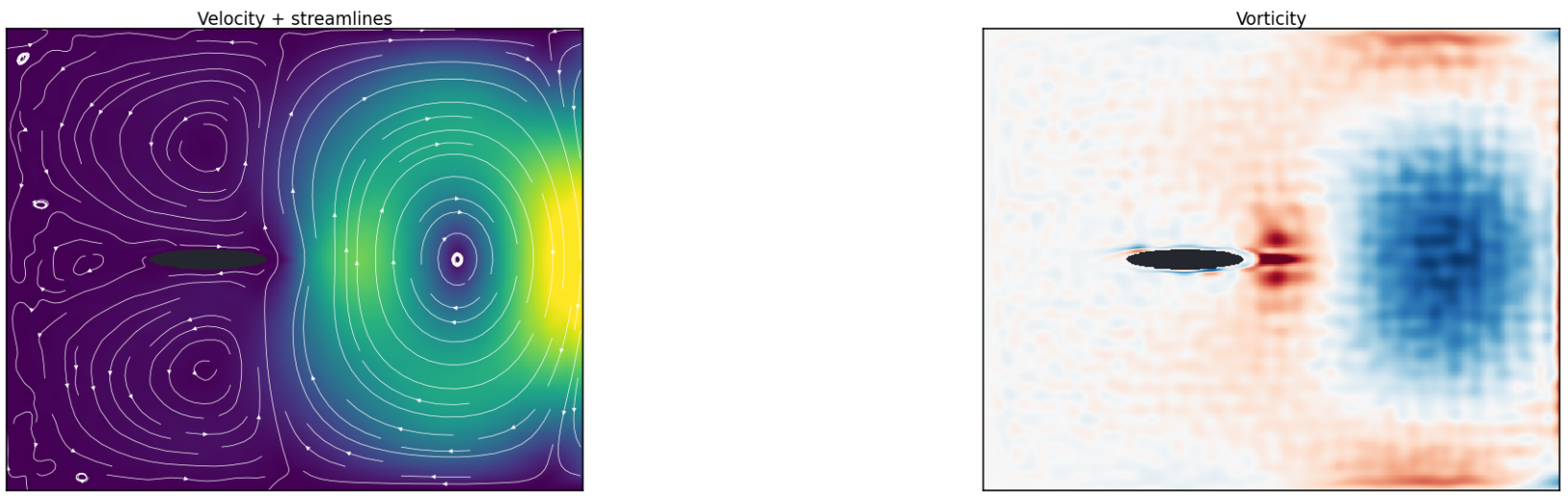
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 2, $\lambda = 9.43449e-01$



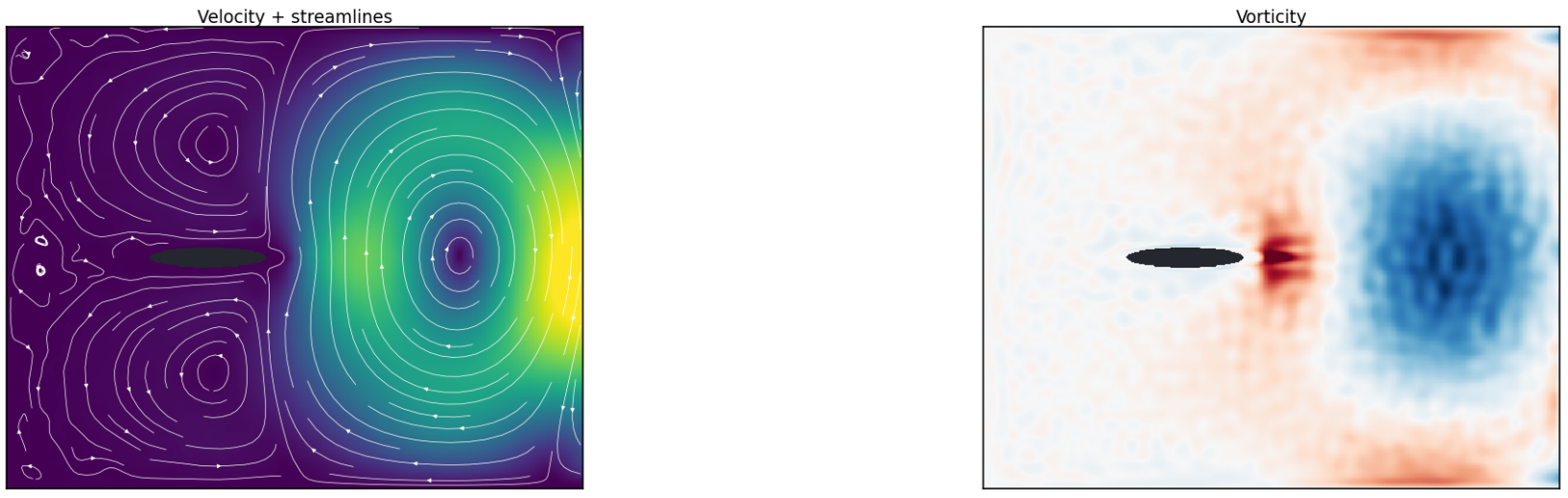
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 2, $\lambda = 9.49746e-01$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

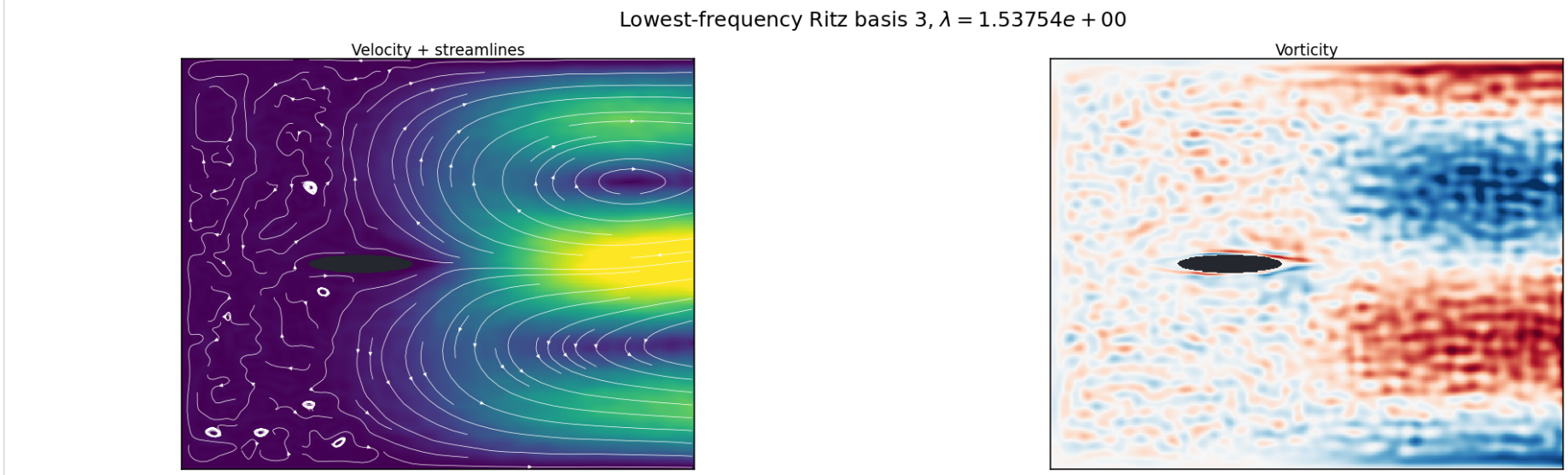
Lowest-frequency Ritz basis 2, $\lambda = 9.33541e-01$



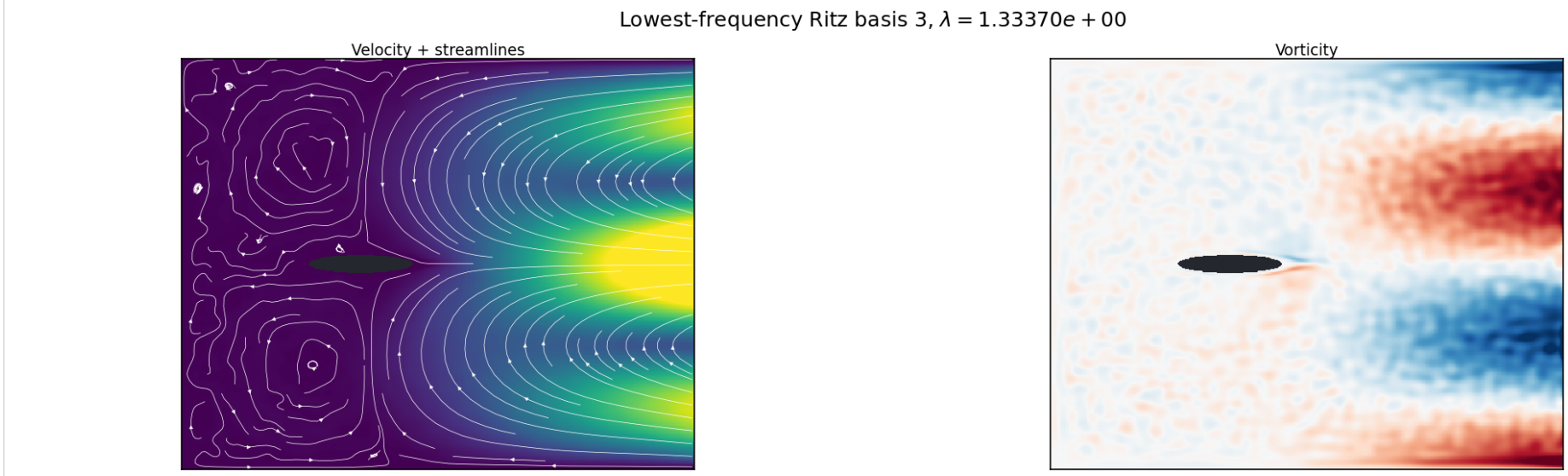
Middle frame | generalized Ritz mode 3 of 6

beta=0.85, alpha=0 deg | left: velocity + streamlines | right: vorticity

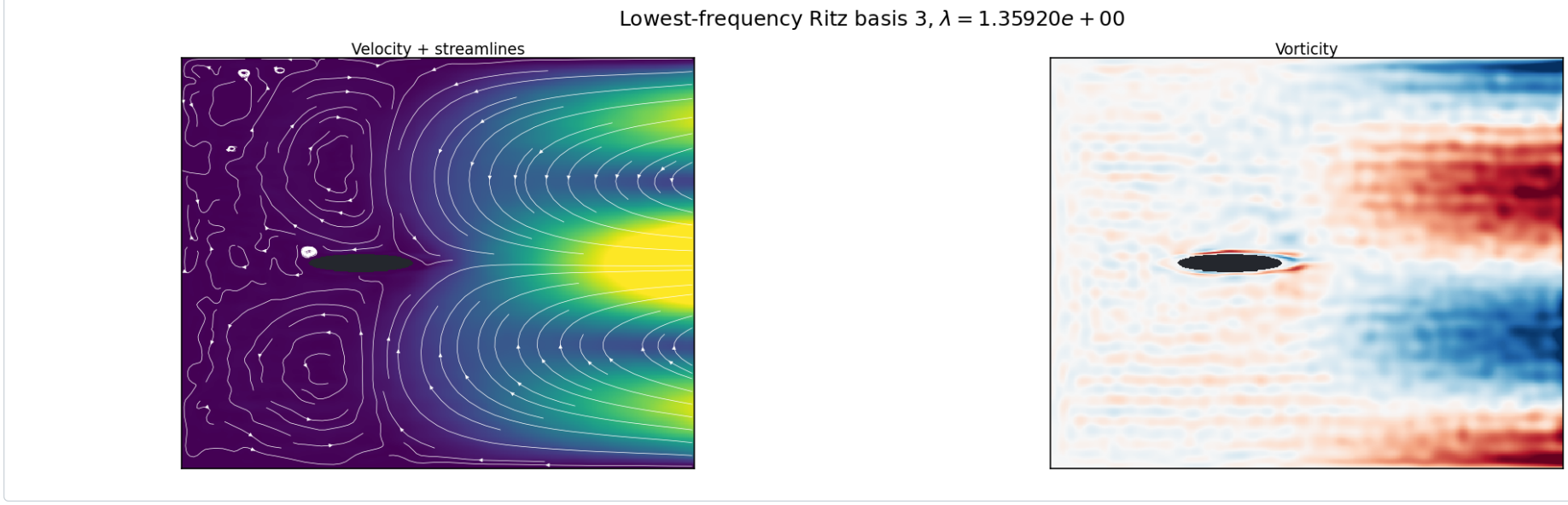
FP16 V100 | epoch 1000 | FP16 velocity 100%



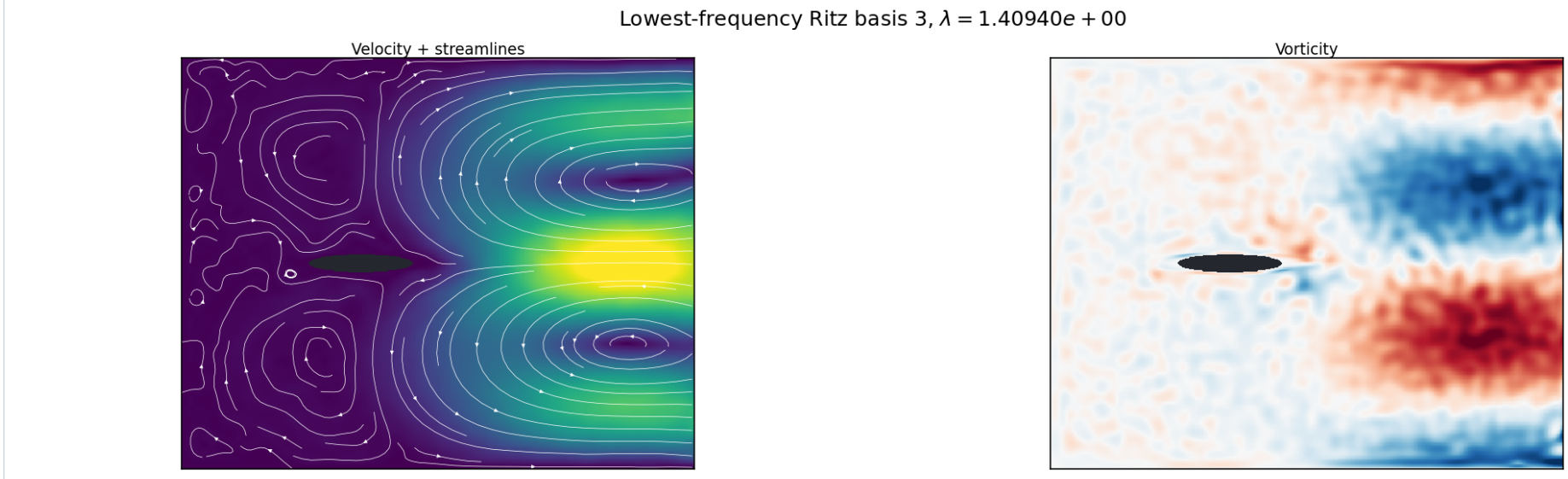
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%



FP64 V100 | epoch 5000 | FP64 velocity 100%



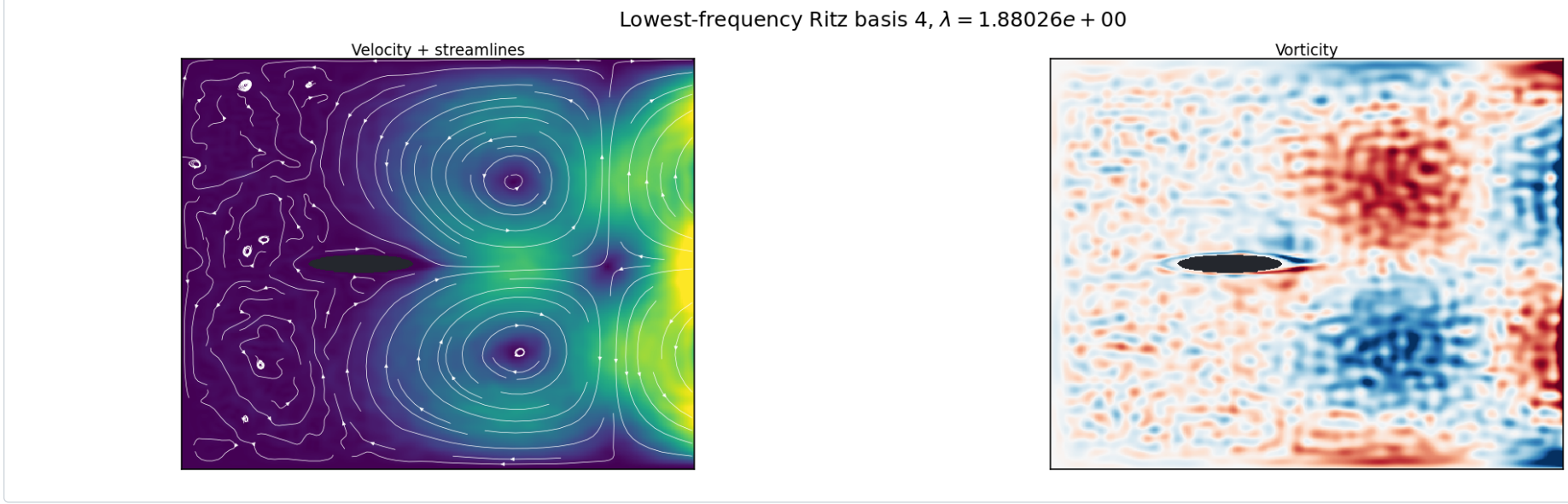
FP16 W100 | epoch 5000 | FP16 vorticity 100%



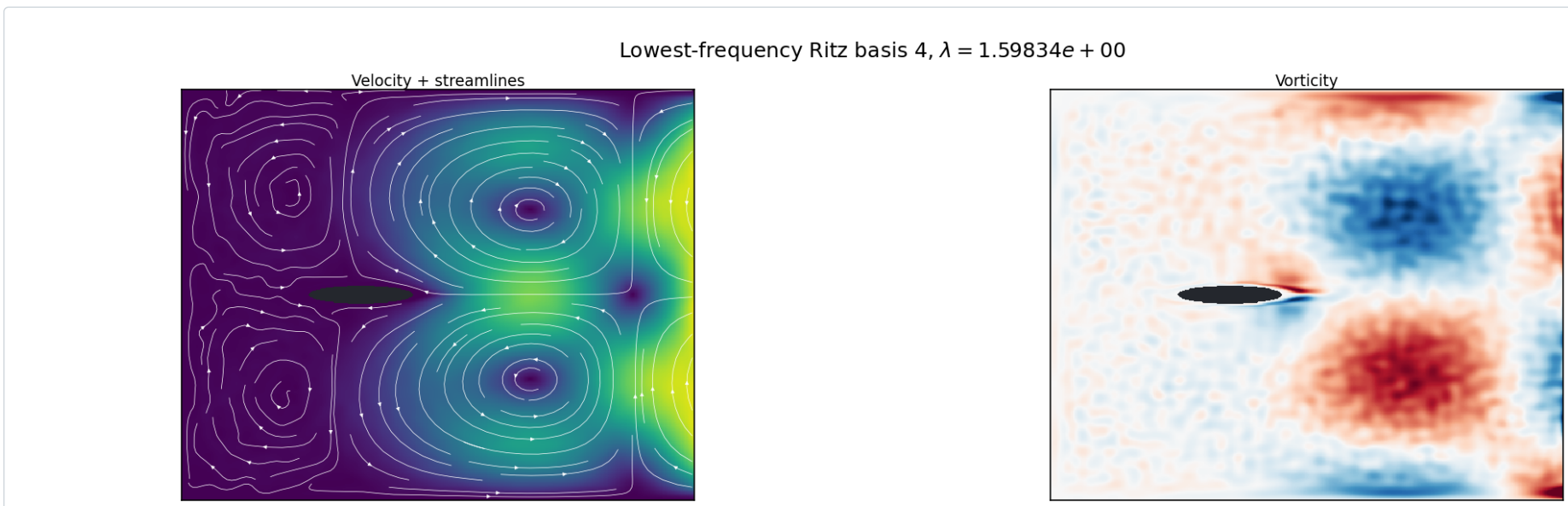
Middle frame | generalized Ritz mode 4 of 6

beta=0.85, alpha=0 deg | left: velocity + streamlines | right: vorticity

FP16 V100 | epoch 1000 | FP16 velocity 100%

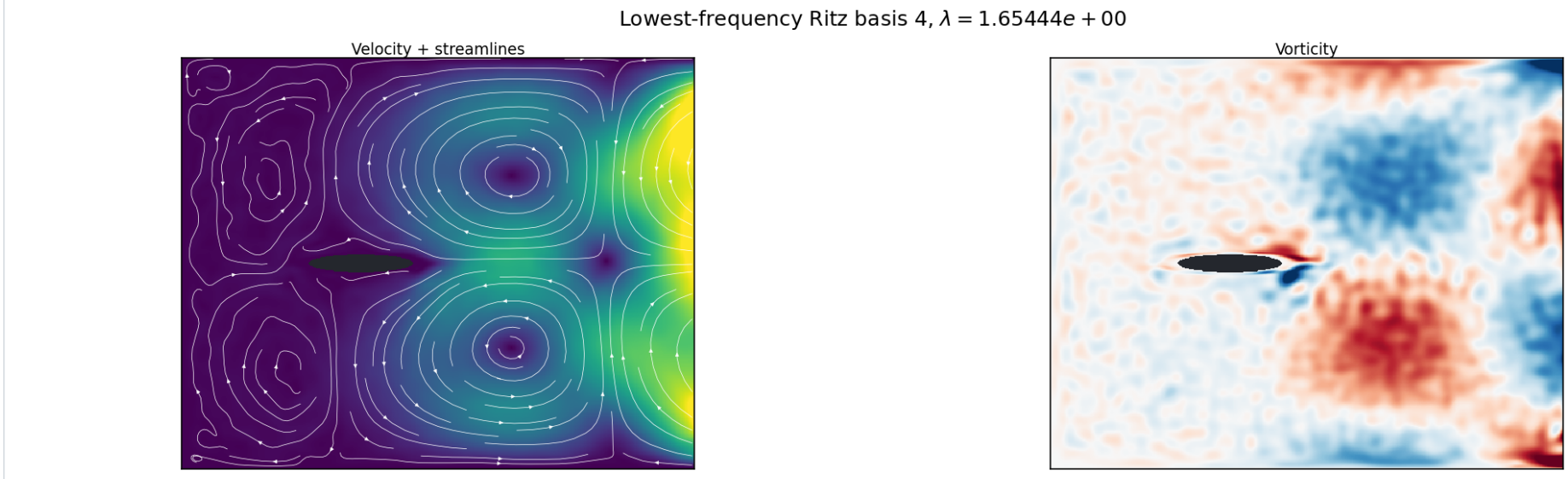


FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%



FP64 V100 | epoch 5000 | FP64 velocity 100%

FP16 W100 | epoch 5000 | FP16 vorticity 100%

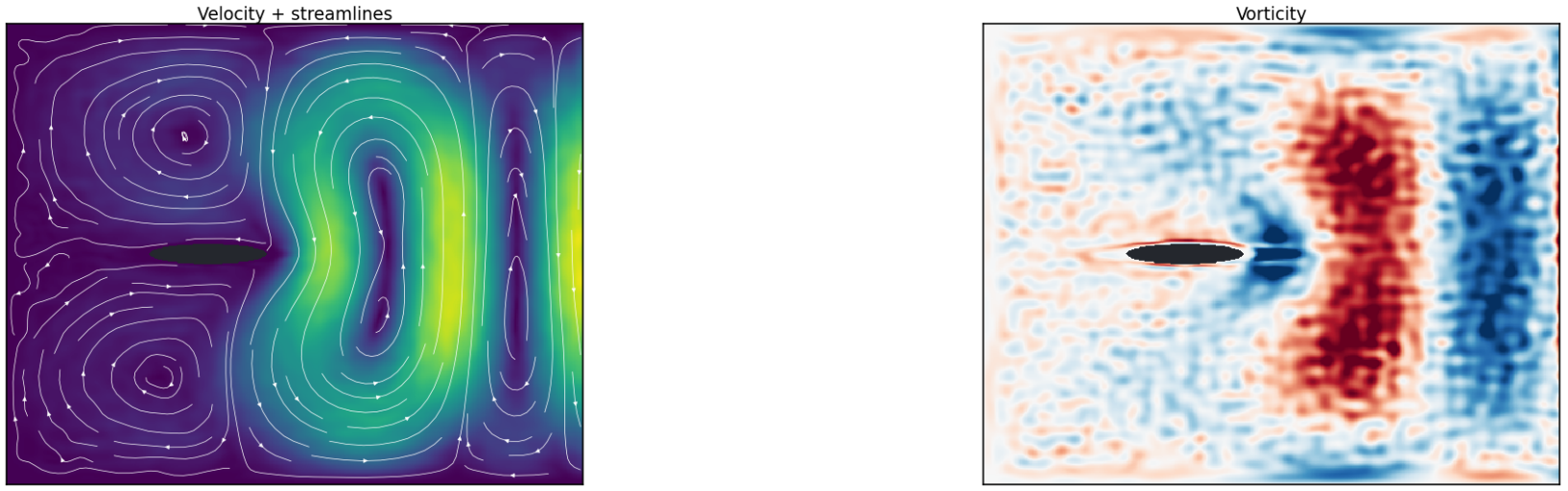


Middle frame | generalized Ritz mode 5 of 6

beta=0.85, alpha=0 deg | left: velocity + streamlines | right: vorticity

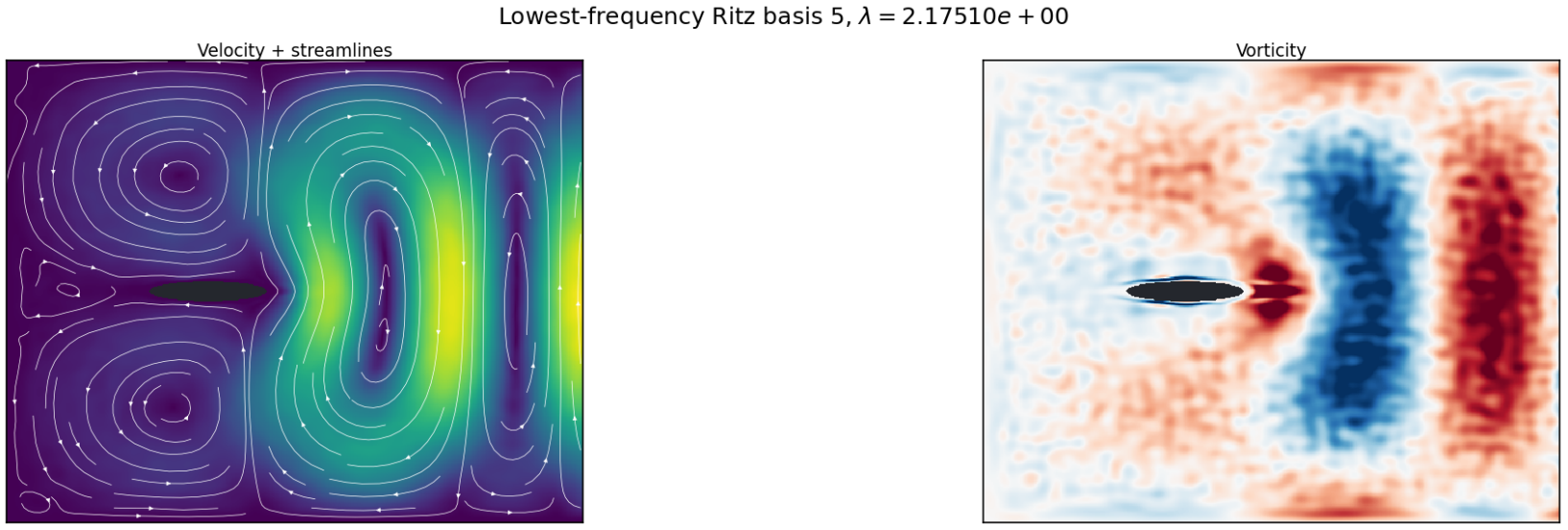
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 5, $\lambda = 2.34874e+00$



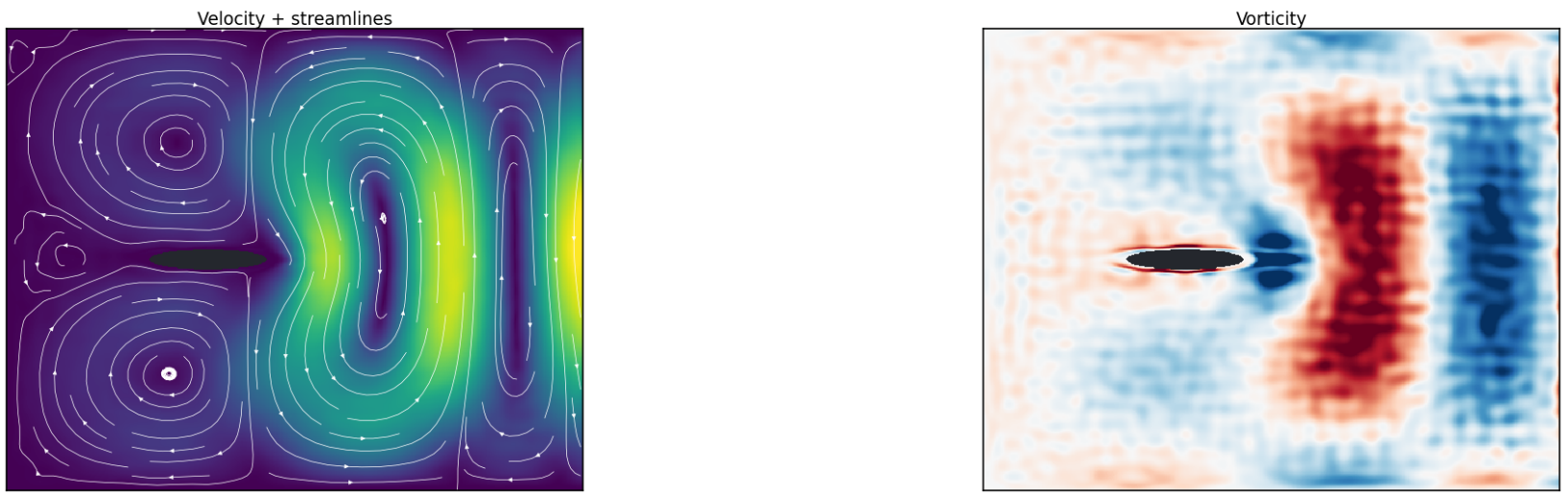
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 5, $\lambda = 2.17510e+00$



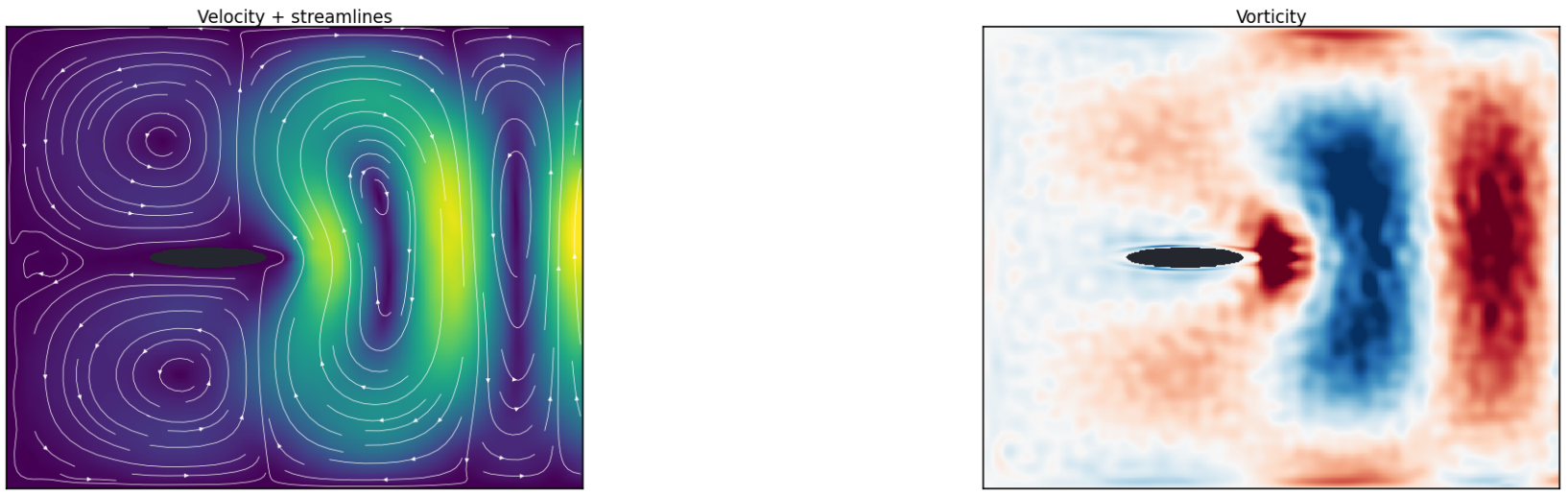
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 5, $\lambda = 2.19582e+00$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 5, $\lambda = 2.17101e+00$

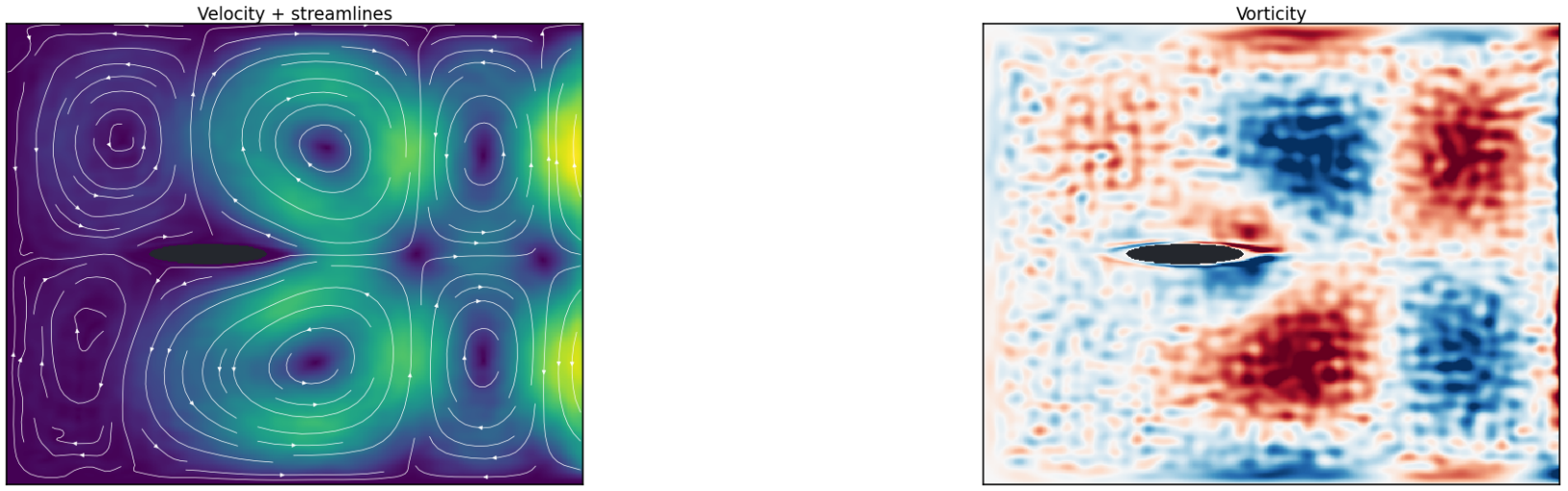


Middle frame | generalized Ritz mode 6 of 6

beta=0.85, alpha=0 deg | left: velocity + streamlines | right: vorticity

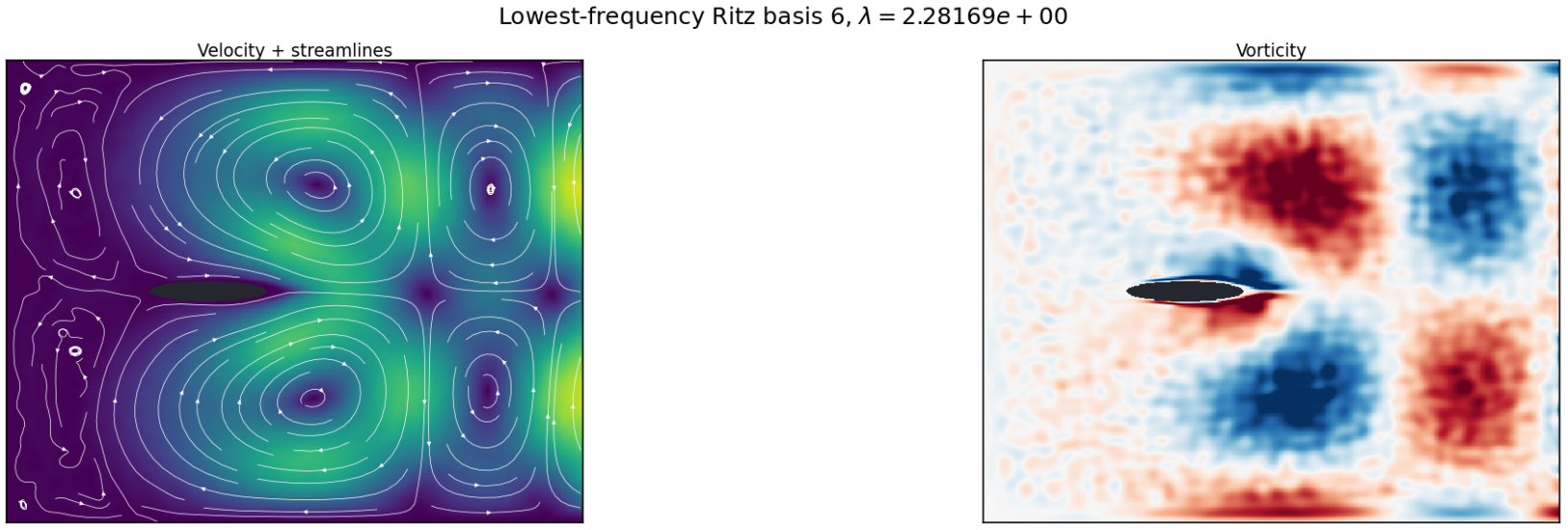
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 6, $\lambda = 2.52172e+00$



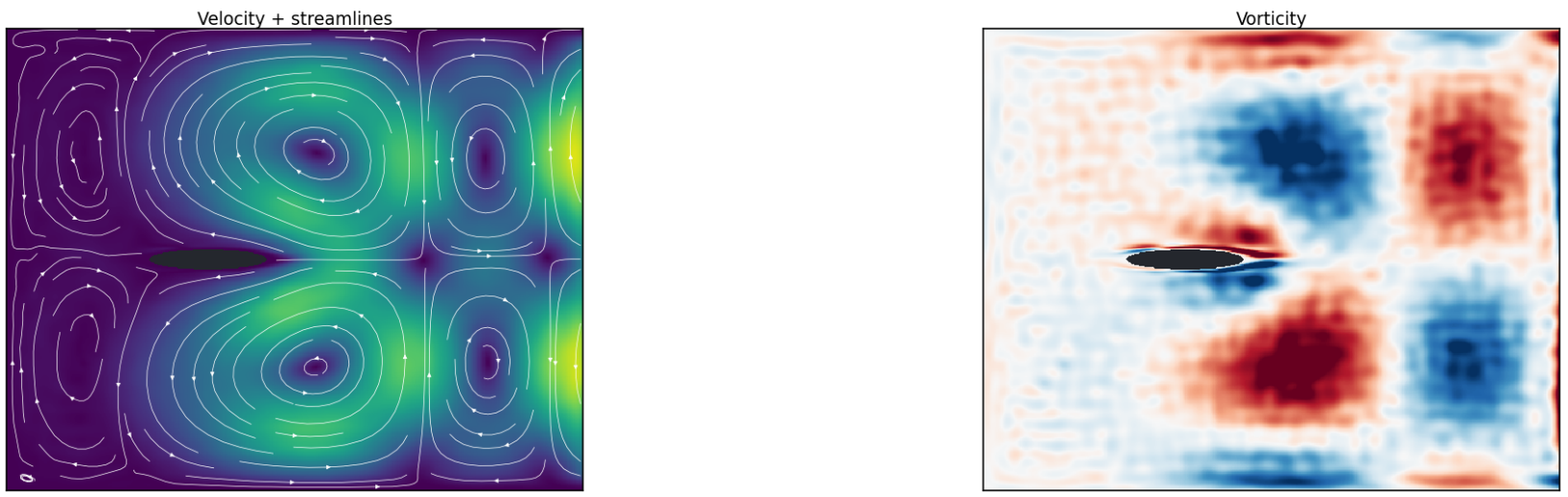
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 6, $\lambda = 2.28169e+00$



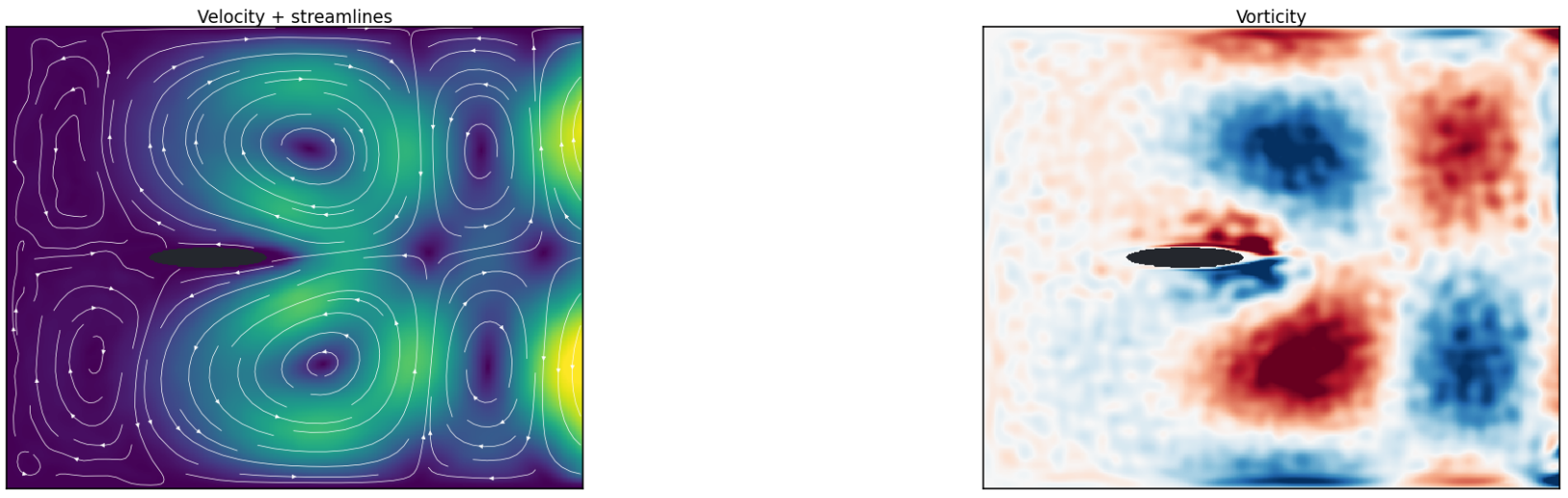
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 6, $\lambda = 2.32233e+00$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 6, $\lambda = 2.30261e+00$

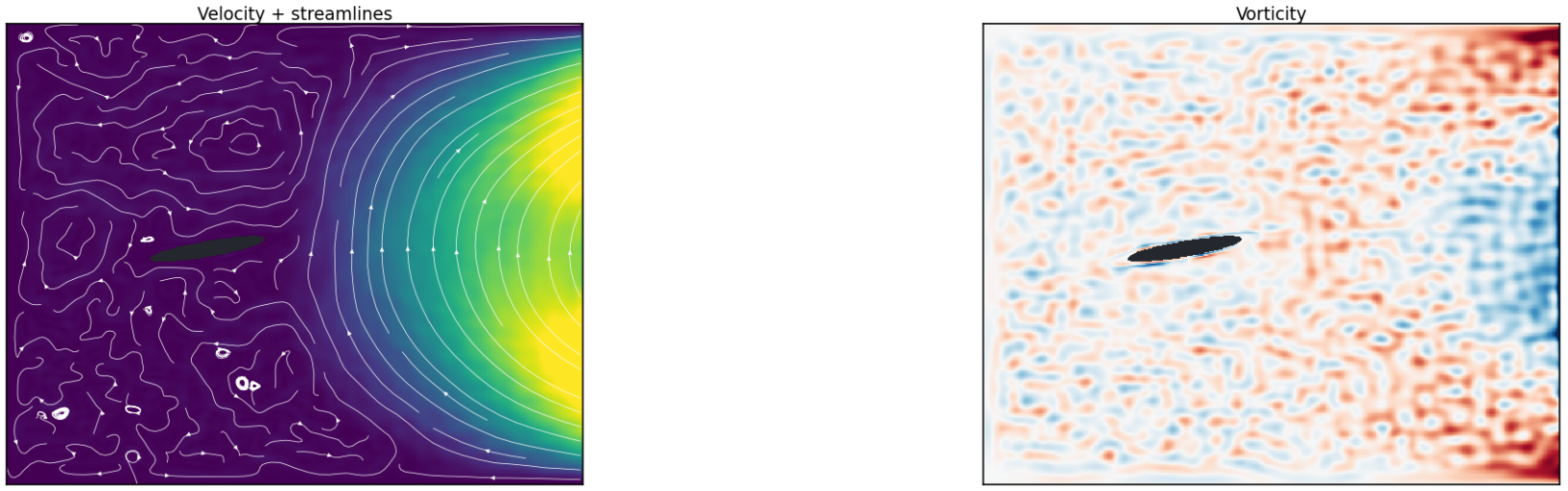


Lower frame | generalized Ritz mode 1 of 6

beta=0.50, alpha=-10 deg | left: velocity + streamlines | right: vorticity

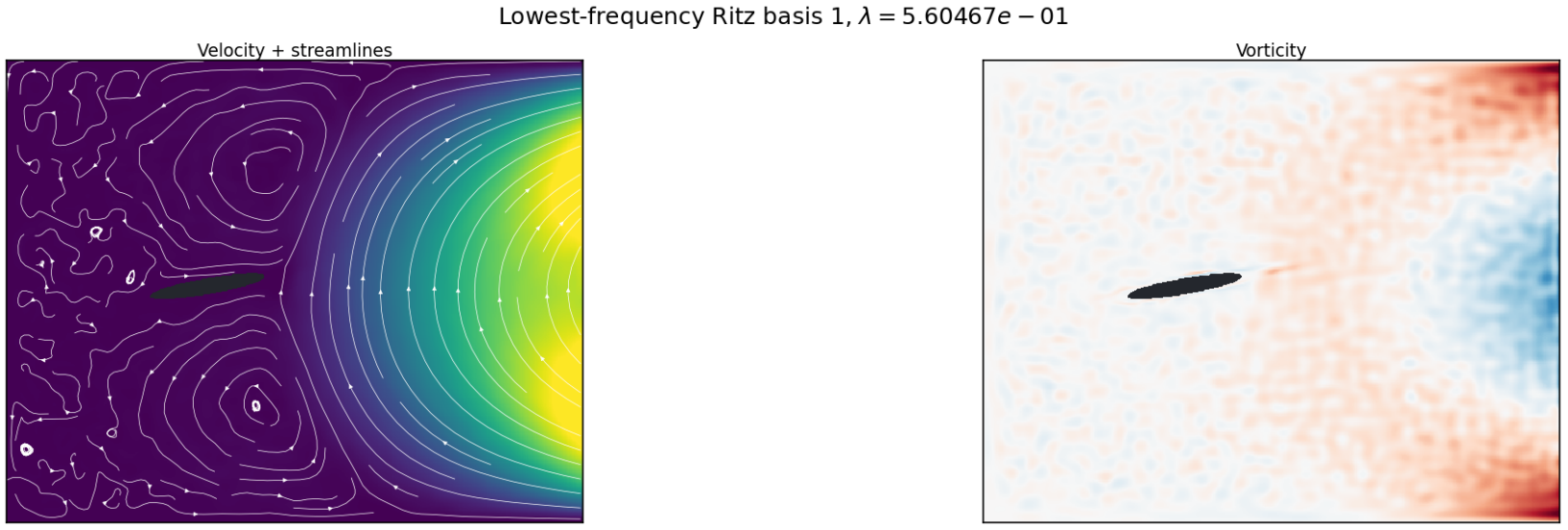
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 1, $\lambda = 8.47617e-01$



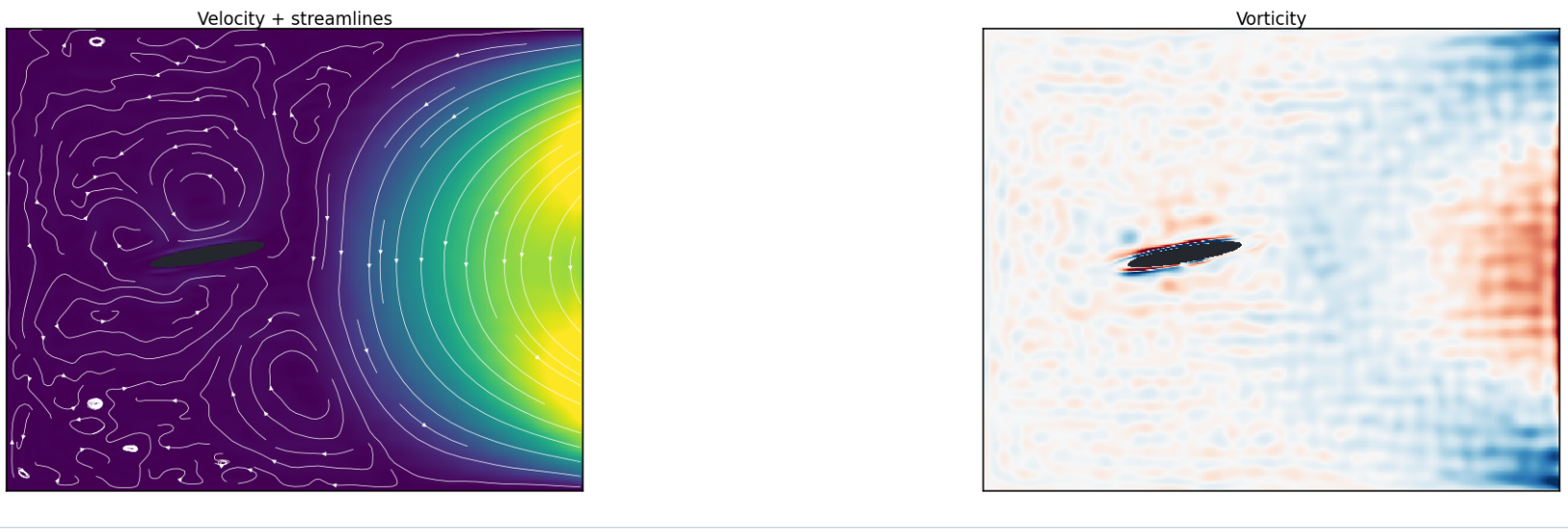
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 1, $\lambda = 5.60467e-01$



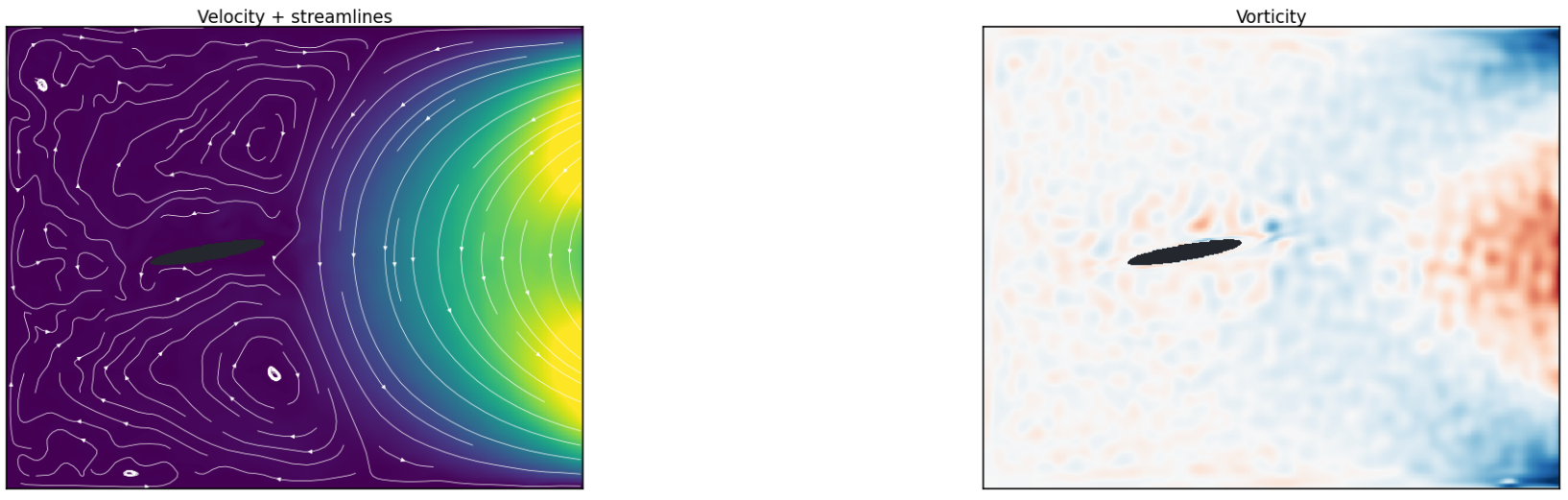
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 1, $\lambda = 6.41310e-01$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 1, $\lambda = 5.80020e-01$

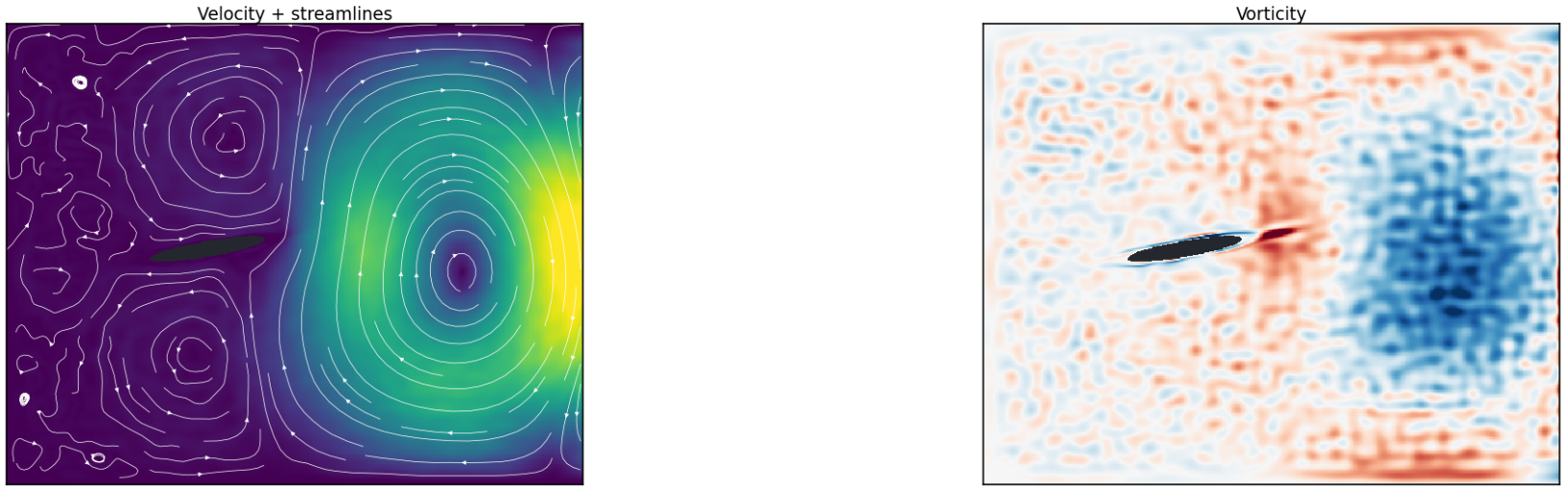


Lower frame | generalized Ritz mode 2 of 6

beta=0.50, alpha=-10 deg | left: velocity + streamlines | right: vorticity

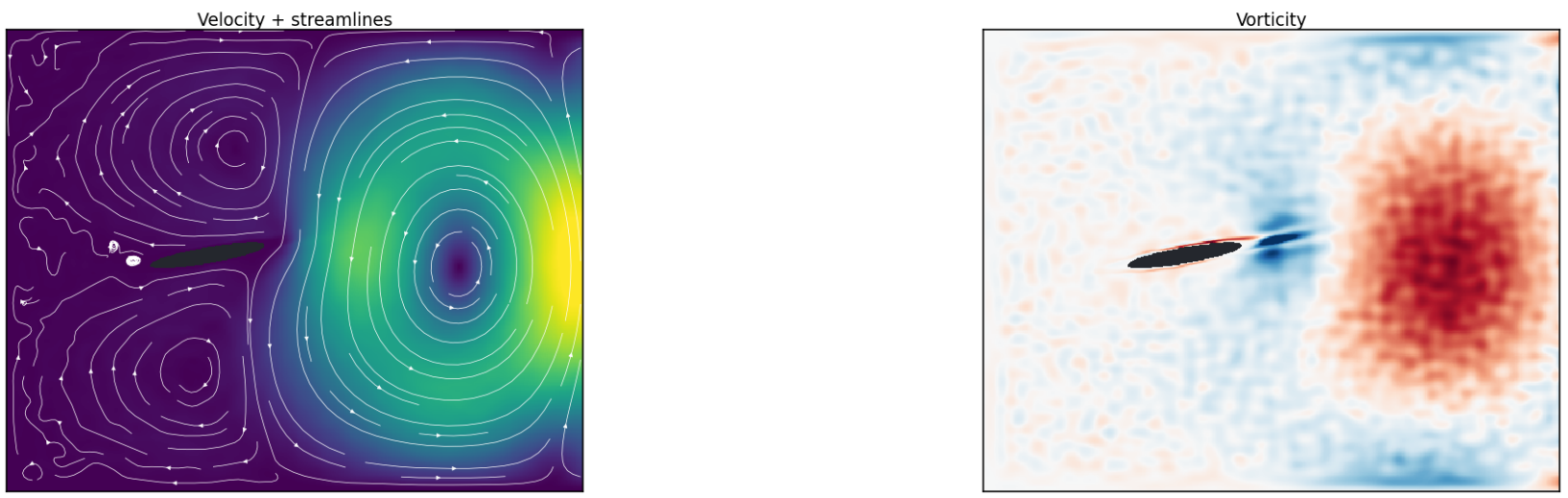
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 2, $\lambda = 1.12627e+00$



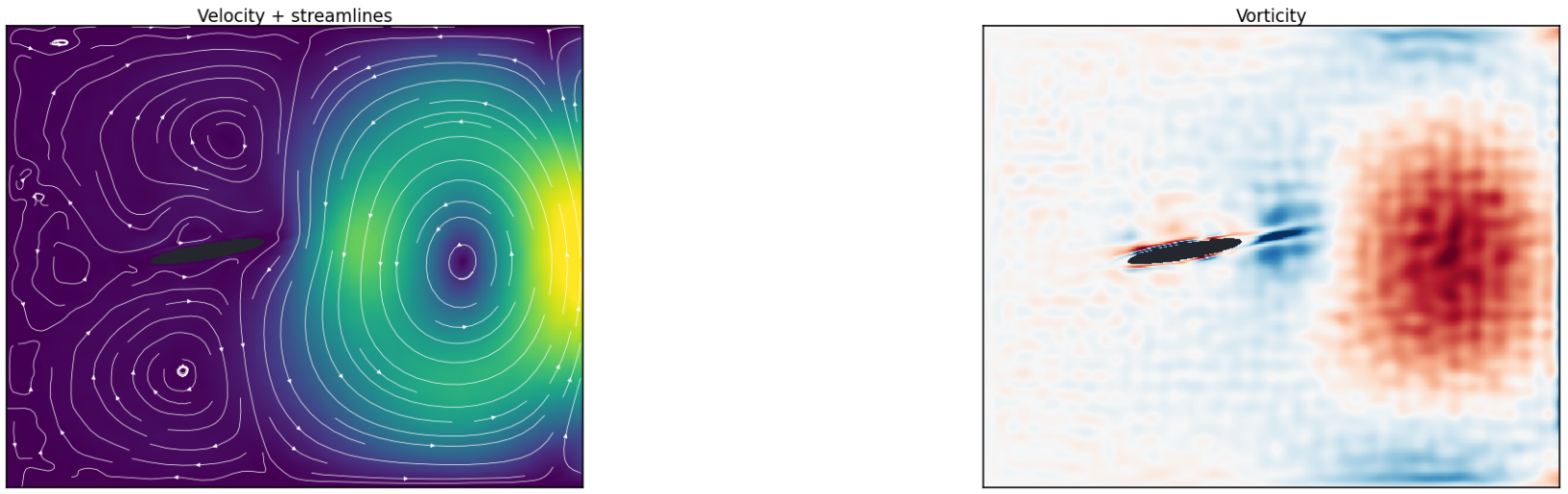
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 2, $\lambda = 9.39716e-01$



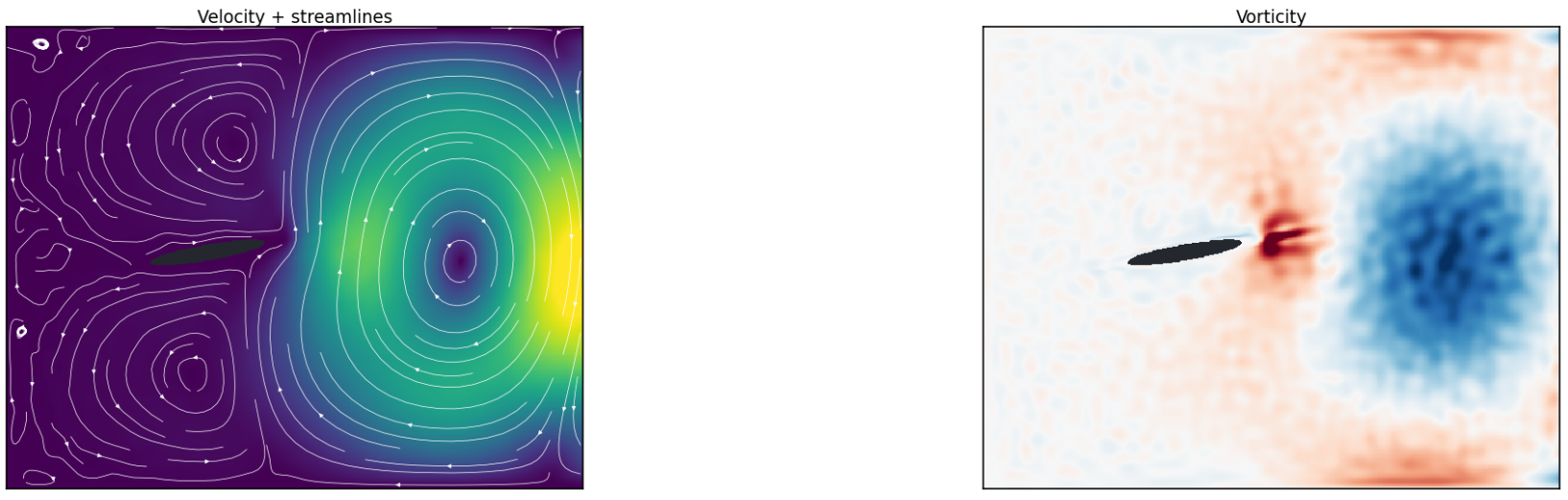
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 2, $\lambda = 9.65967e-01$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 2, $\lambda = 9.29818e-01$

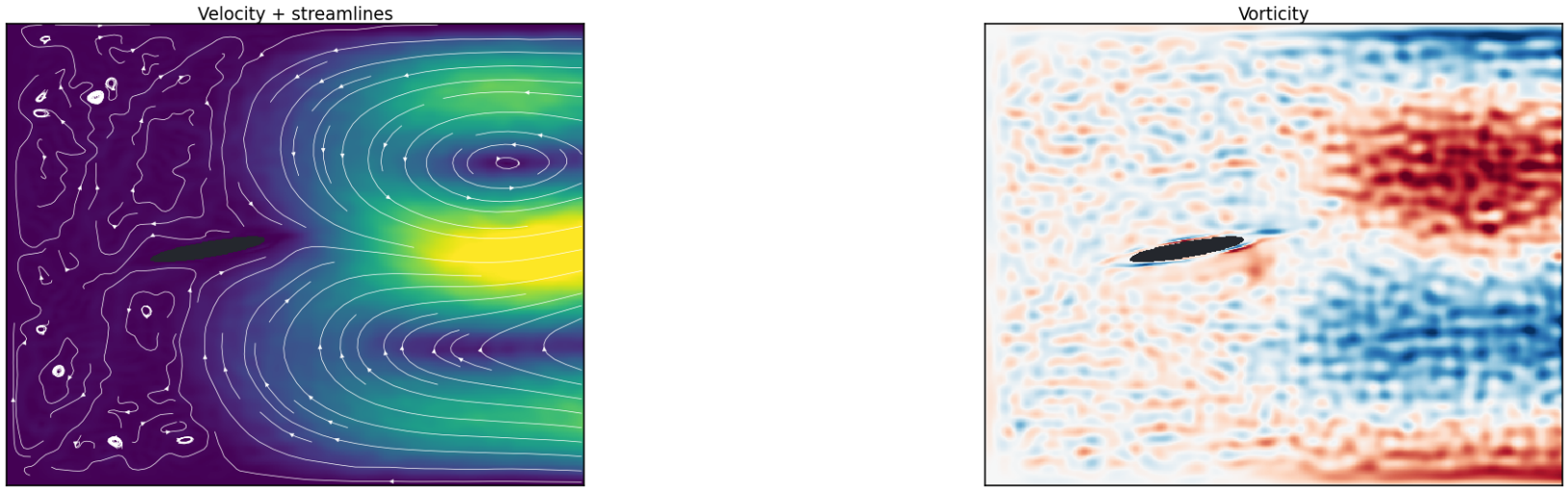


Lower frame | generalized Ritz mode 3 of 6

beta=0.50, alpha=-10 deg | left: velocity + streamlines | right: vorticity

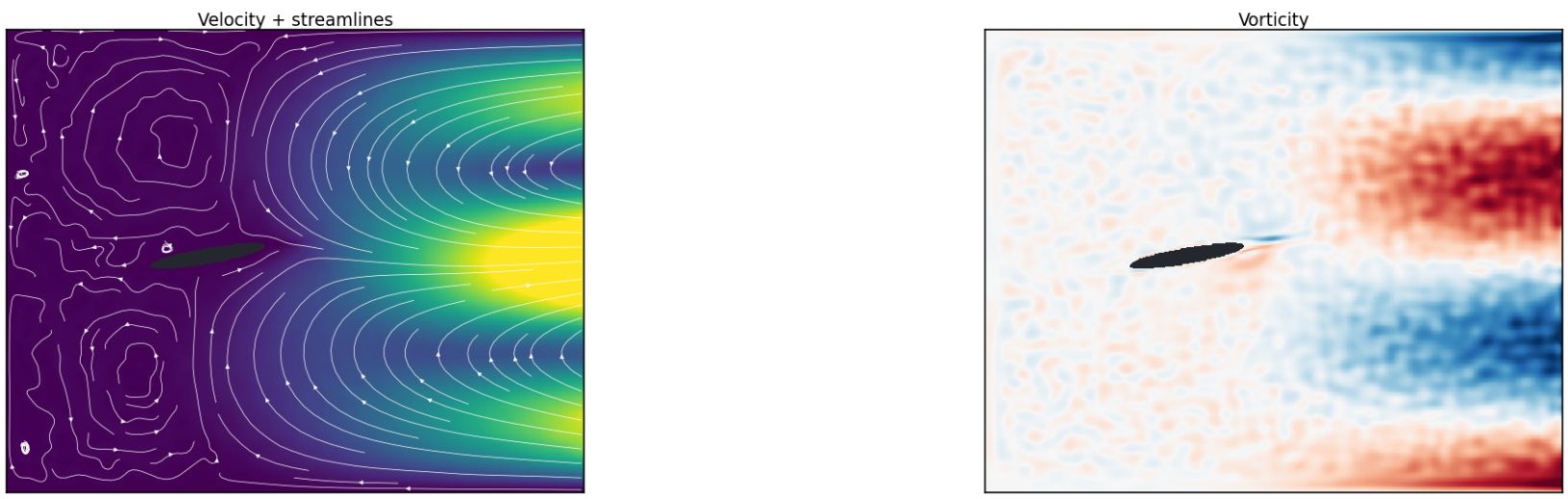
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 3, $\lambda = 1.57887e+00$



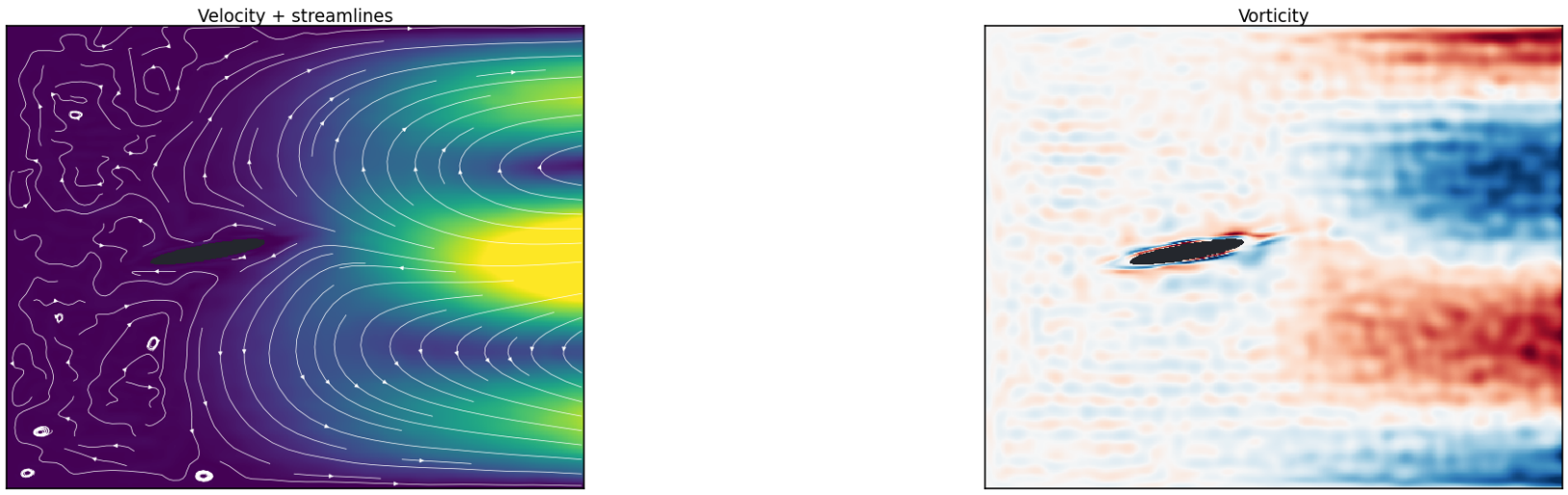
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 3, $\lambda = 1.33139e+00$



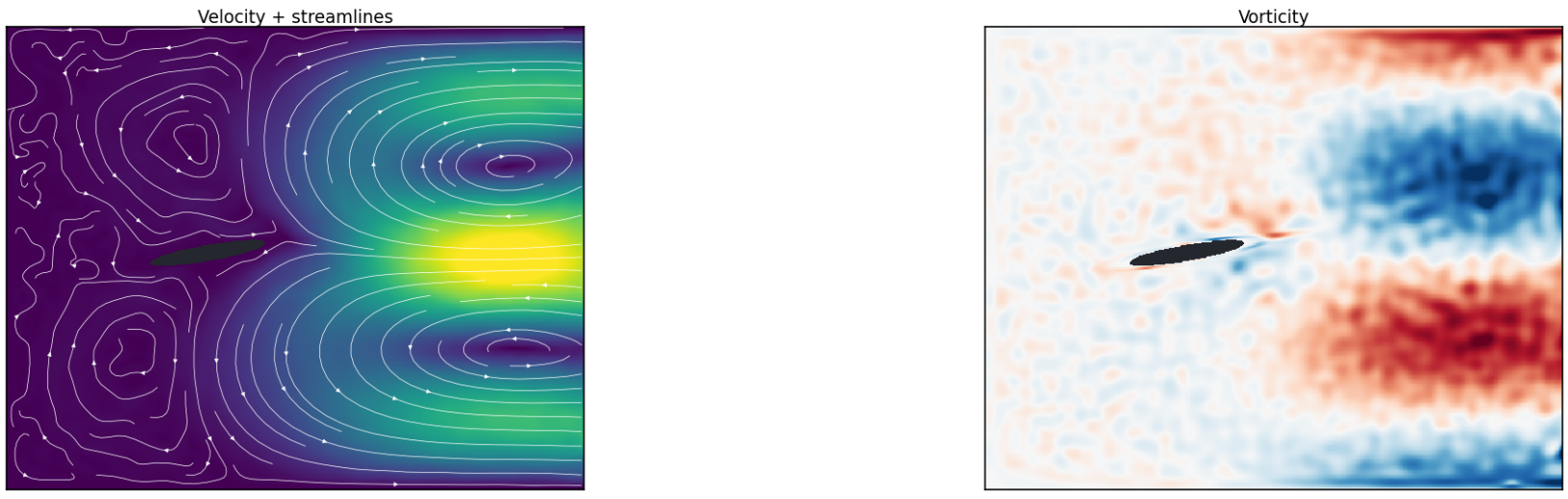
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 3, $\lambda = 1.38753e+00$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 3, $\lambda = 1.38974e+00$

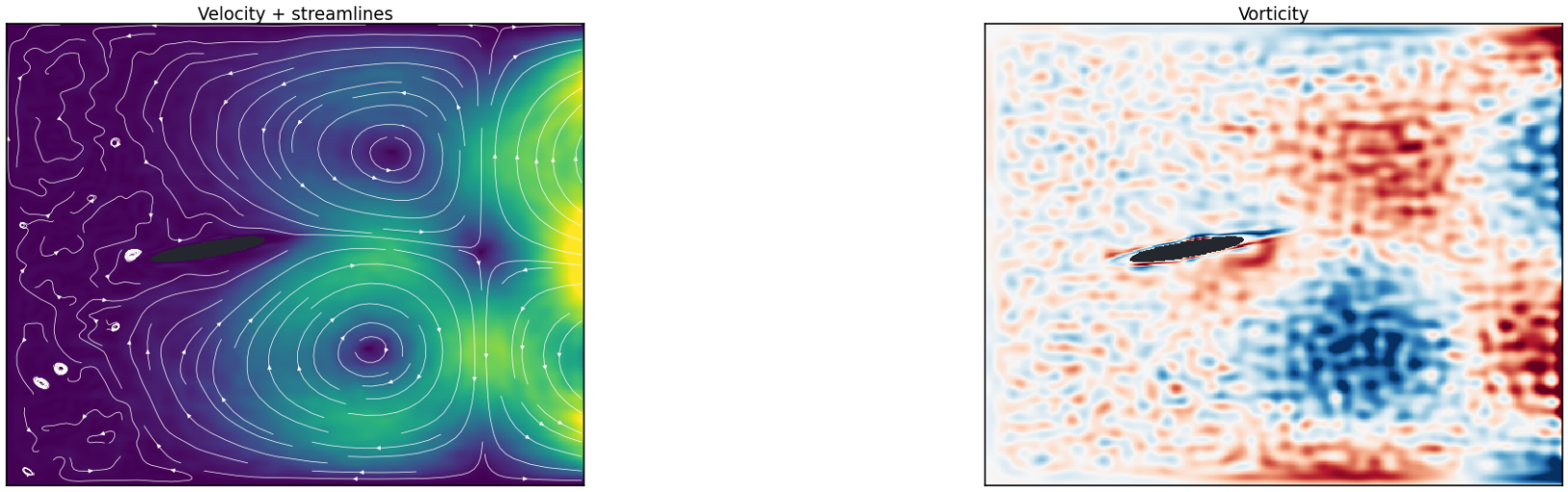


Lower frame | generalized Ritz mode 4 of 6

beta=0.50, alpha=-10 deg | left: velocity + streamlines | right: vorticity

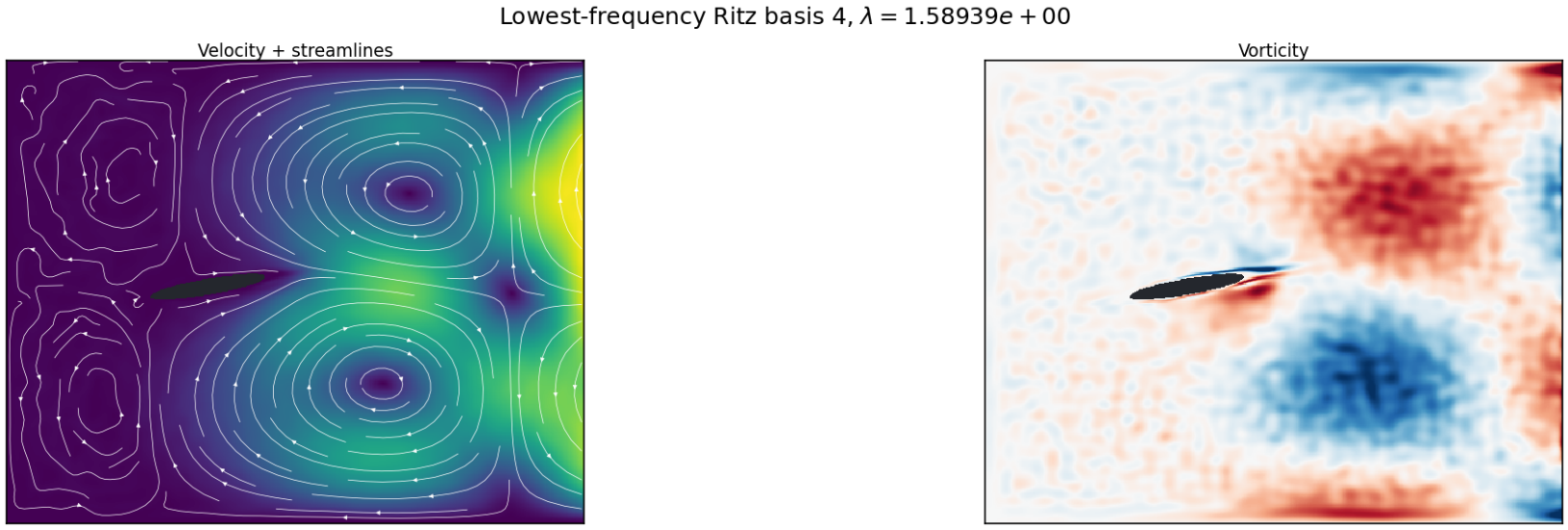
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 4, $\lambda = 1.95559e+00$



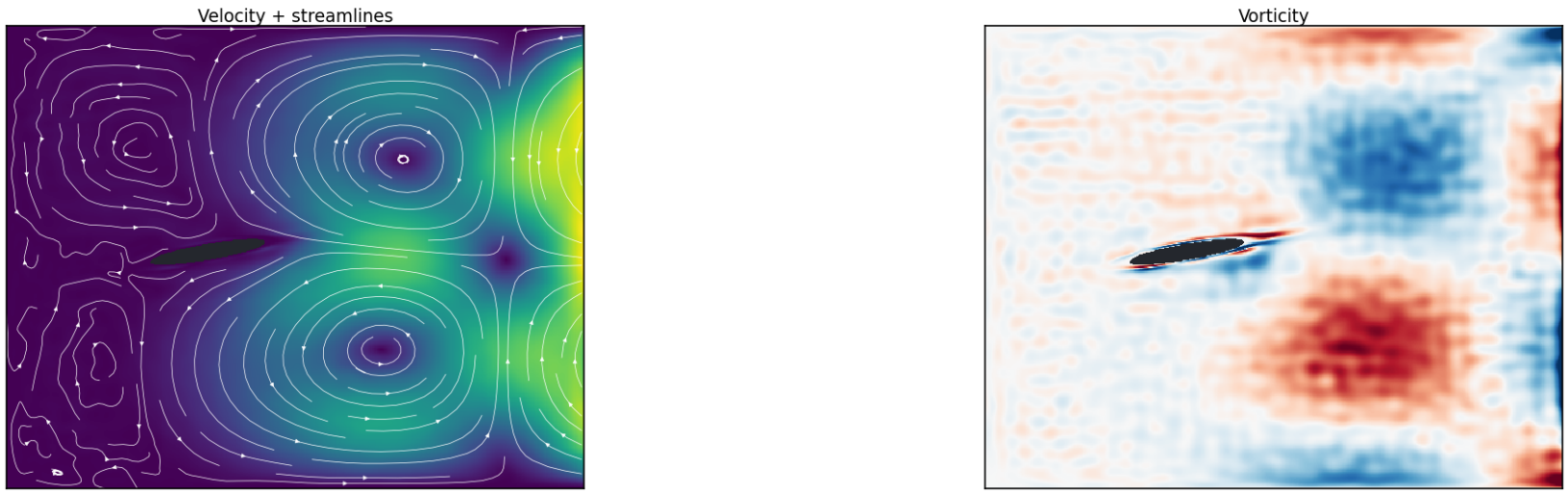
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 4, $\lambda = 1.58939e+00$



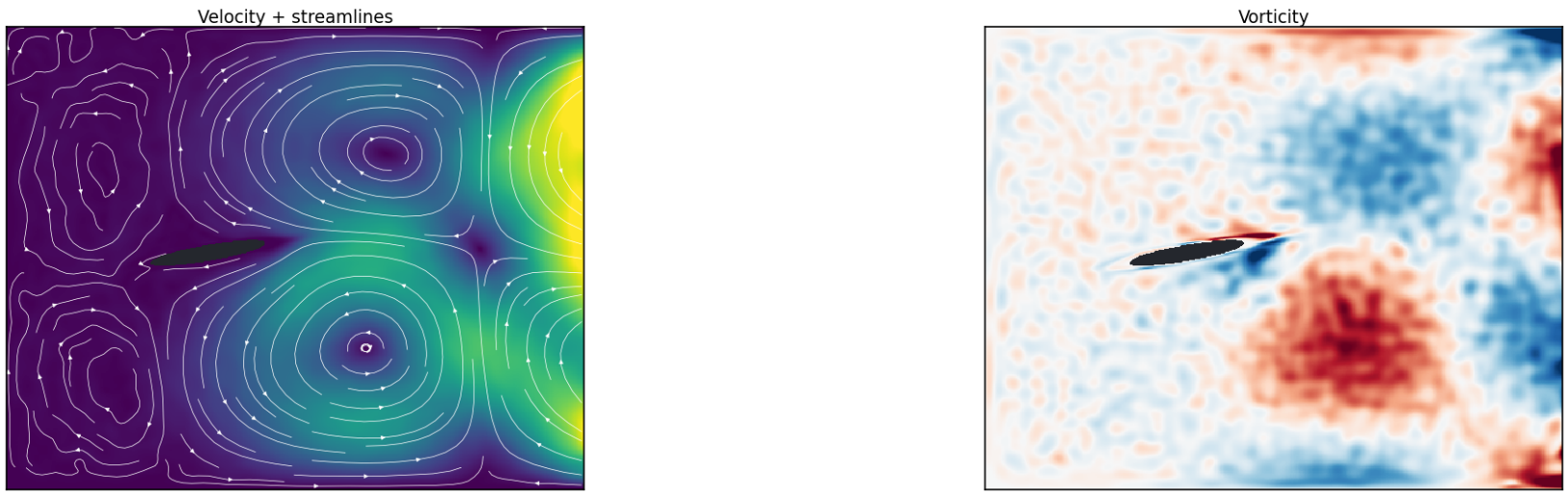
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 4, $\lambda = 1.63111e+00$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 4, $\lambda = 1.66097e+00$



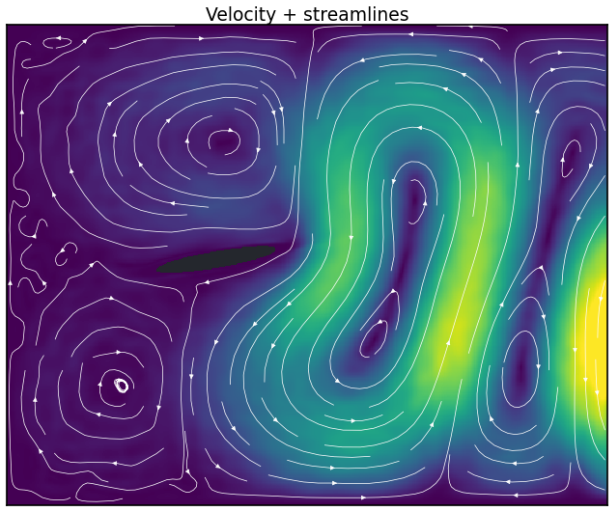
Lower frame | generalized Ritz mode 5 of 6

beta=0.50, alpha=-10 deg | left: velocity + streamlines | right: vorticity

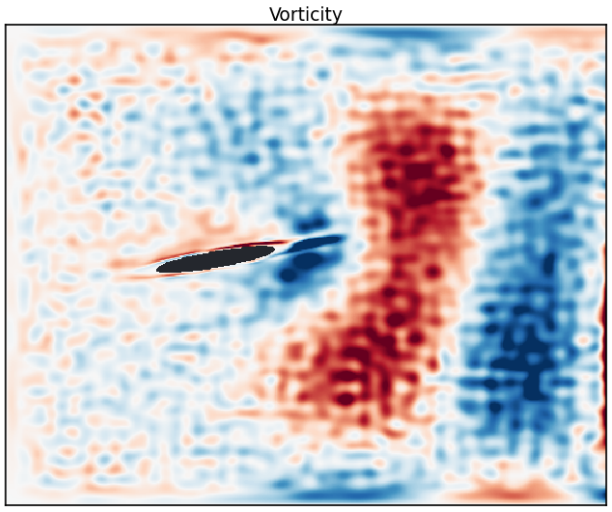
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 5, $\lambda = 2.32736e+00$

Velocity + streamlines



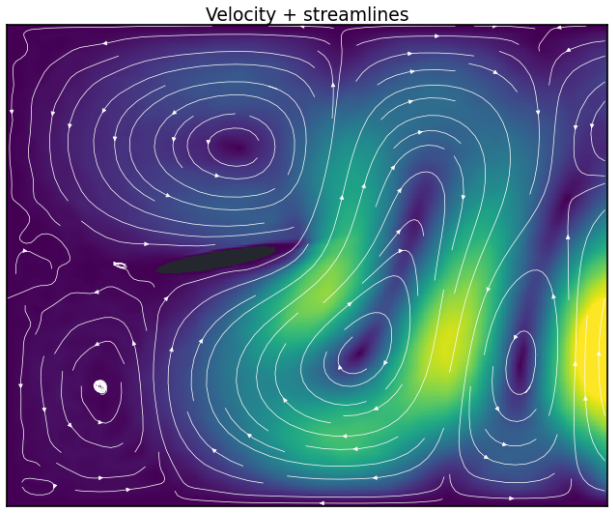
Vorticity



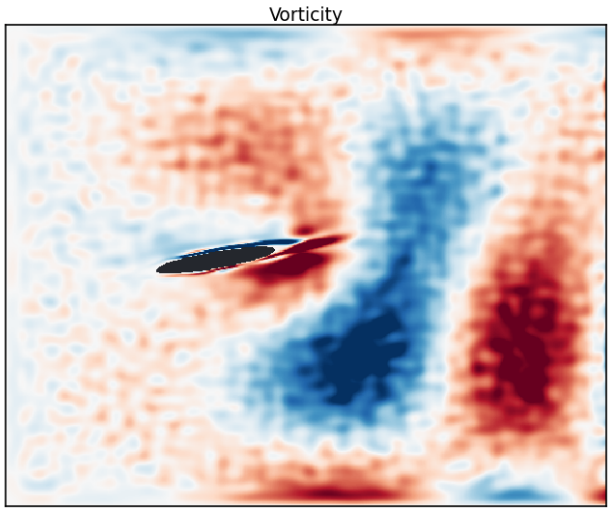
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 5, $\lambda = 2.09645e+00$

Velocity + streamlines



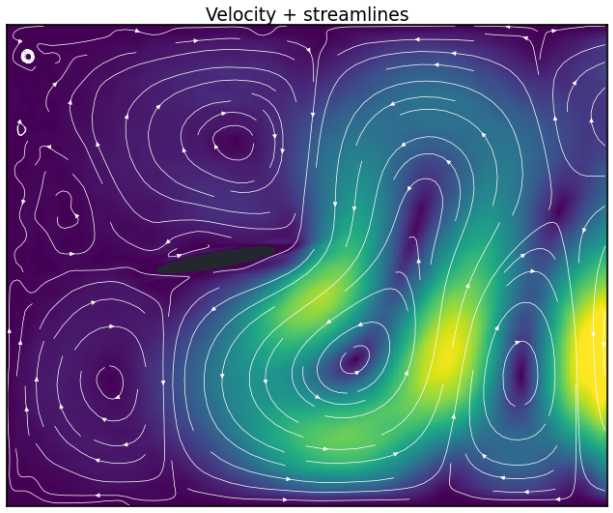
Vorticity



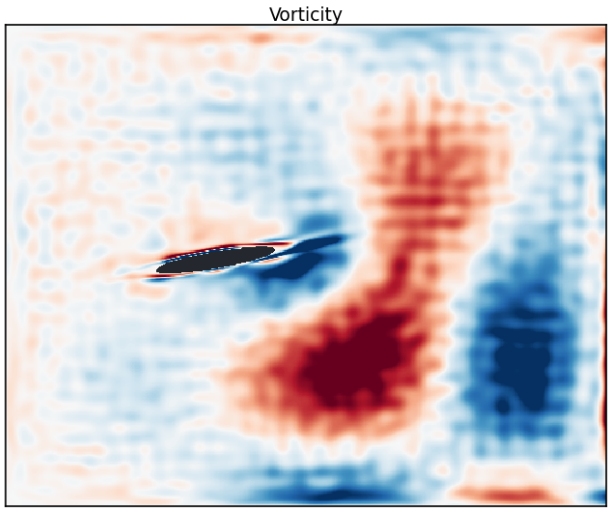
FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 5, $\lambda = 2.18016e+00$

Velocity + streamlines



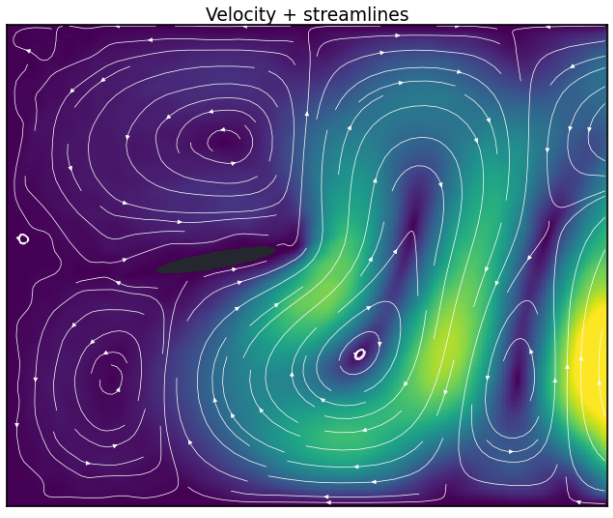
Vorticity



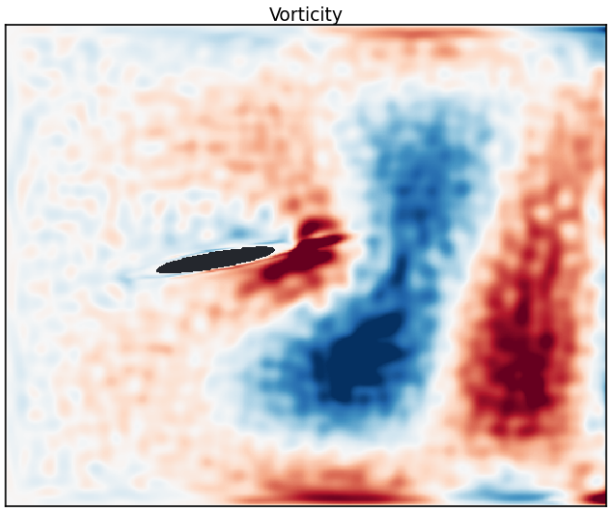
FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 5, $\lambda = 2.11859e+00$

Velocity + streamlines



Vorticity

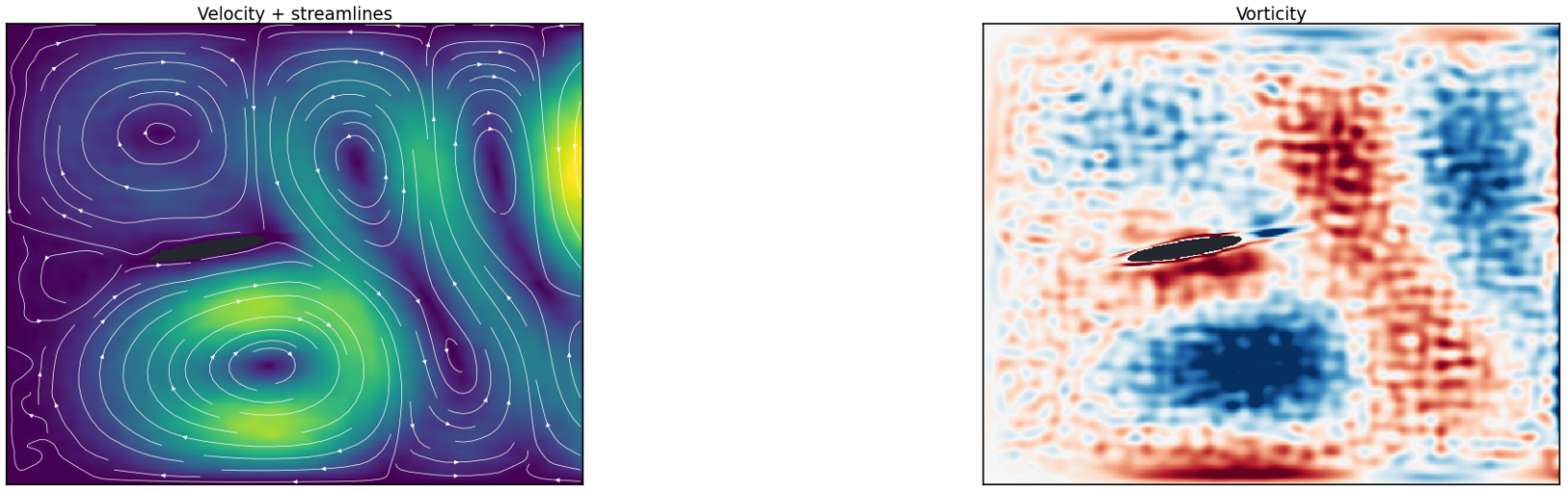


Lower frame | generalized Ritz mode 6 of 6

beta=0.50, alpha=-10 deg | left: velocity + streamlines | right: vorticity

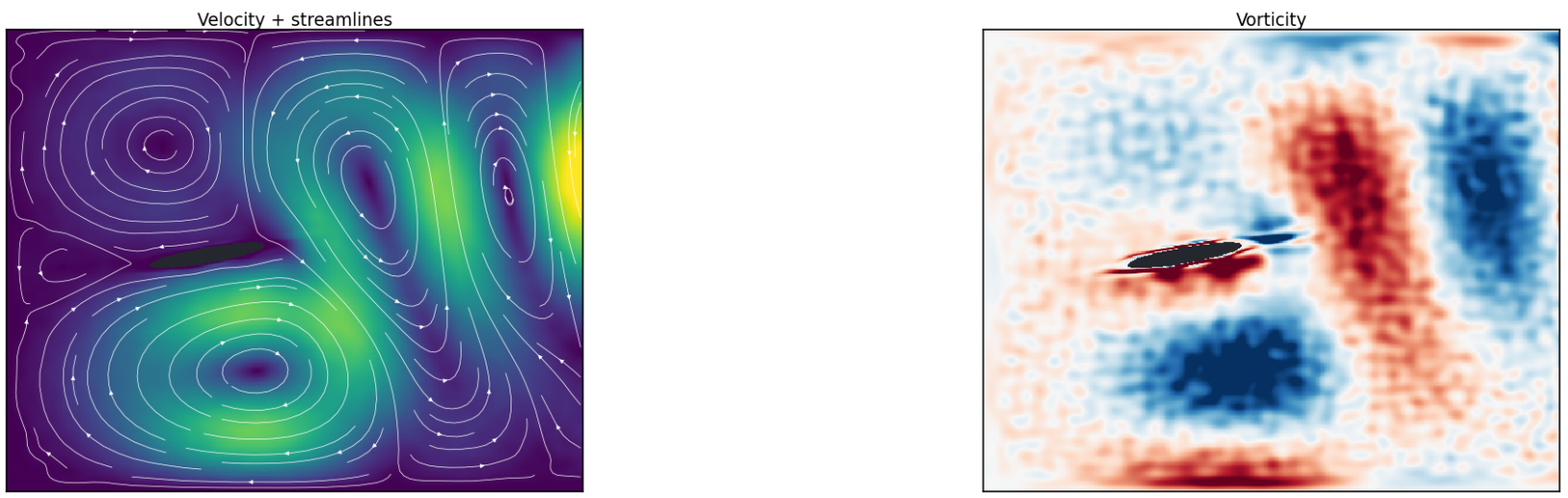
FP16 V100 | epoch 1000 | FP16 velocity 100%

Lowest-frequency Ritz basis 6, $\lambda = 2.59535e+00$



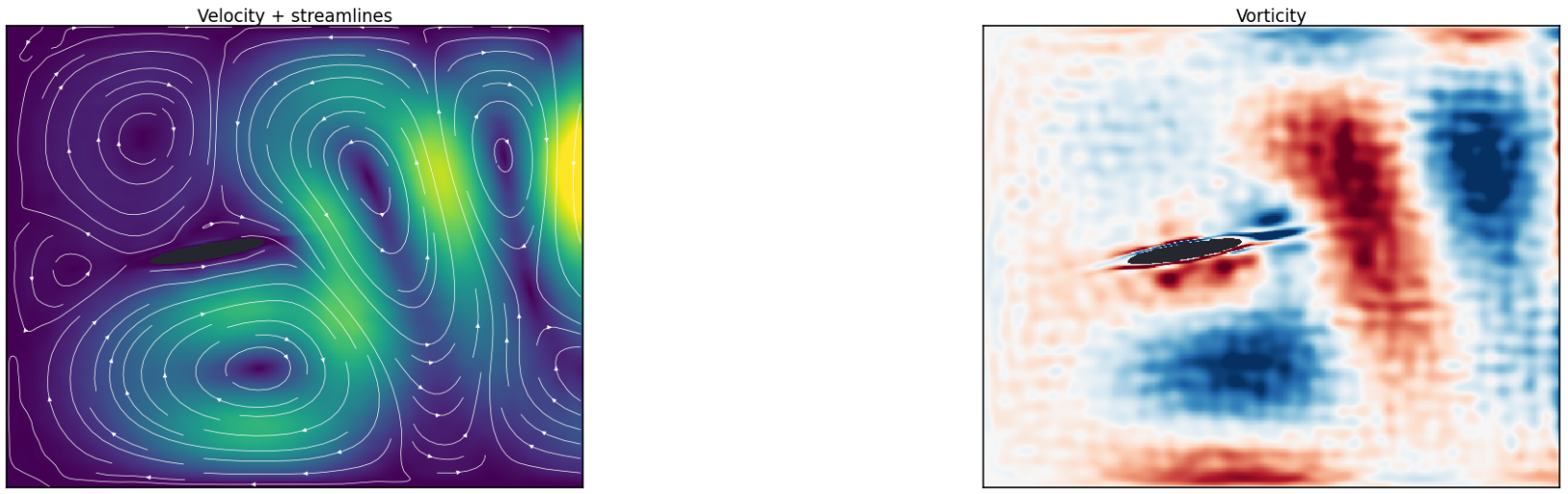
FP16 V50/W50 | epoch 650 | FP16 velocity 50% + vorticity 50%

Lowest-frequency Ritz basis 6, $\lambda = 2.29104e+00$



FP64 V100 | epoch 5000 | FP64 velocity 100%

Lowest-frequency Ritz basis 6, $\lambda = 2.36166e+00$



FP16 W100 | epoch 5000 | FP16 vorticity 100%

Lowest-frequency Ritz basis 6, $\lambda = 2.28296e+00$